Report

on

Vayu

Real Time Air Quality Monitor

Submitted To:

University Institute of Computing

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BONAFIDE CERTIFICATE

Certified that this project report "Vayu Real Time Air Quality Monitor" is the Bonafide work of "Pragya Patel (UID 24MCI10008)" who carried out the project work under my/our supervision.

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Submitted for the project viva-voce examination

INTERNAL EXAMINER

EXTERNAL EXAMIN

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ABSTRACT

Air quality degradation has emerged as one of the most pressing environmental and public health concerns in contemporary India. Rapid urbanisation, industrial expansion, vehicular proliferation, and agricultural burning have collectively led to hazardous concentrations of airborne pollutants in major cities. The Air Quality Index (AQI) is the primary scientific metric endorsed by India's Central Pollution Control Board (CPCB) to communicate ambient air quality and its associated health implications to the general public, policymakers, and the research community.

This report presents VAYU, a real-time, web-based AQI prediction and analysis system developed using Python and the Streamlit framework. The application integrates live environmental data from the OpenWeatherMap REST APIs — including the Geocoding API, the Air Pollution API, and the 5-day Forecast API — to retrieve current concentrations of seven key pollutants: PM2.5, PM10, NO_2, SO_2, CO, O_3, and NH_3. The AQI is computed strictly following CPCB's official sub-index linear interpolation formula, where the final AQI is determined as the maximum sub-index across all measured pollutants.

VAYU provides an interactive and visually rich dashboard featuring an animated Plotly gauge meter, pollutant-wise sub-index bar charts, a pollutant radar map, and a 5-day AQI forecast trend line. Beyond visualisation, the system delivers structured health intelligence including category-specific health impact summaries, recommended daily precautions, and differentiated risk assessments for four demographic groups — children, adults, the elderly, and individuals with pre-existing conditions. A one-click PDF report generation feature enables users to download a formatted academic report suitable for institutional submission. The system is evaluated for technical feasibility, data accuracy relative to official CPCB advisories, and user experience, demonstrating its potential as an academic tool, a civic awareness platform, and a research prototype.

Keywords: Air Quality Index, AQI Prediction, CPCB Standard, Streamlit, Plotly, OpenWeatherMap, PM2.5, PM10, Health Intelligence, Environmental Data Science, Real-Time Monitoring, Python, FPDF2, Data Visualisation

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CHAPTER 1
INTRODUCTION

Air quality degradation has emerged as one of the most pressing environmental and public health concerns in contemporary India. Rapid urbanisation, industrial expansion, vehicular proliferation, and agricultural burning have collectively led to hazardous concentrations of airborne pollutants in major cities. The Air Quality Index (AQI) is the primary scientific metric endorsed by India's Central Pollution Control Board (CPCB) to communicate ambient air quality and its associated health implications to the general public, policymakers, and the research community.

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1.1 Need Identification

Air pollution has emerged as one of the defining environmental crises of the 21st century, with India bearing a disproportionately high burden of its consequences. The World Health Organisation (WHO) estimates that outdoor air pollution contributes to approximately 4.2 million premature deaths globally each year, with the South Asian subcontinent accounting for a significant share. India's rapid economic development — characterised by accelerating urbanisation, industrial growth, expanding road transport, and continued agricultural burning practices — has created a complex, multi-source pollution landscape in which ambient concentrations of key pollutants frequently exceed both national and international safety thresholds.

The Central Pollution Control Board (CPCB) of India introduced the National Air Quality Index in 2014 to provide a standardised, single-number metric communicating the overall state of ambient air quality at a given location and time. The AQI system uses a 0–500 scale divided into six health-based categories — Good, Satisfactory, Moderate, Poor, Very Poor, and Severe — each associated with specific health implications and advisory guidance. Despite its scientific robustness and official endorsement, the AQI remains poorly understood by the general public, primarily because the tools available for accessing and interpreting AQI data are either too technical for non-specialist use or insufficiently comprehensive for academic and research purposes.

From an academic perspective, the development of an AQI analysis system represents an exceptional opportunity to integrate competencies from multiple engineering and scientific disciplines within a single coherent project: REST API integration, environmental data science, standards-compliant algorithmic computation, interactive data visualisation, public health informatics, user interface design, and automated document generation. Such integrative projects are highly valued in undergraduate engineering education as they demonstrate the ability to synthesise diverse technical capabilities into functional, real-world solutions.

The motivation for VAYU specifically arises from the observation that despite the maturity of open APIs for environmental data (notably OpenWeatherMap's Air Pollution API), no open-source Python tool exists that performs complete CPCB-compliant AQI computation, provides multi-dimensional interactive visualisation, delivers structured health intelligence, and generates formatted academic reports — all within a single accessible web interface requiring no specialist technical knowledge to operate.

VAYU fills this gap by providing a comprehensive, accessible, and academically rigorous AQI analysis platform that serves students, researchers, environmental engineers, public health workers, and engaged citizens with a common interface meeting the needs of each audience.

1.2 Identification of Problem

A systematic analysis of the current AQI information landscape in India reveals several distinct problems that motivated the design and development of VAYU:

1.2.1 Fragmentation of Environmental Data Sources

Pollutant concentration data, meteorological conditions, geolocation information, and historical air quality records reside across multiple APIs, databases, and government platforms that are not natively integrated for individual city-level analysis. The CPCB's AQI portal, while authoritative, does not expose its raw data via a public REST API accessible to developers. Commercial APIs such as OpenWeatherMap and IQAir provide accessible data but require integration effort. This fragmentation means that any comprehensive AQI analysis requires accessing and harmonising data from multiple sources, a task beyond the capabilities of non-technical users.

1.2.2 Complexity of CPCB Computation

The CPCB AQI methodology involves pollutant-specific breakpoint tables and a piecewise linear interpolation formula that, while mathematically straightforward, is non-trivial to implement correctly for all seven regulated pollutants with appropriate unit conversions, boundary condition handling, and dominant pollutant identification. Errors in breakpoint table encoding or formula implementation can produce AQI values that are formally incorrect but difficult to detect without specialist verification. There is a clear need for a reference implementation that is openly available and verifiably accurate.

1.2.3 Absence of Contextualised Health Guidance

Raw AQI numbers, even when accurately computed, are insufficient for informed decision-making without category-specific health advisories. The health implications of an AQI of 180 (Moderate) versus 220 (Poor) differ significantly, and the appropriate protective measures vary further by demographic group — what constitutes acceptable exposure for a healthy adult may be dangerous for a child with asthma or an elderly person with cardiovascular disease. No existing open-source AQI tool combines real-time AQI computation with demographically stratified health guidance in a single interface.

1.2.4 Lack of Academic Documentation Tools

Students and researchers working with AQI data for academic projects, environmental impact assessments, or public health research require formatted, citable reports that document their findings. Existing AQI platforms provide interactive dashboards but offer no facility for generating formatted, downloadable reports suitable for institutional submission or academic documentation.

1.2.5 Accessibility and Interpretability Gap

There exists a substantial gap between the technical sophistication of available environmental data and the interpretability of that data for non-specialist audiences. A system that transforms complex API responses, technical concentration values, and multi-parameter AQI computations into a visually clear, intuitively navigable interface with plain-language health guidance addresses a genuine public health communication need that existing solutions inadequately serve.

1.3 Identification of Tasks

Based on the identified problems, the following specific technical and design tasks were defined for this project:

- Design and implement a Python-based web application using Streamlit that accepts a city name and API key as inputs and delivers a complete real-time AQI analysis dashboard within a single-page interface.
- Integrate OpenWeatherMap's Geocoding API, Air Pollution API (current data), and Air Pollution Forecast API (5-day/3-hour intervals) to retrieve real-time and forecasted environmental data for any searchable Indian city.
- Implement the complete CPCB AQI computation algorithm, including accurate pollutant-specific breakpoint tables for all seven regulated pollutants (PM2.5, PM10, NO2, SO2, CO, O3, NH3), the piecewise linear sub-index interpolation formula, the maximum sub-index rule for final AQI, and dominant pollutant identification.
- Develop four interactive Plotly visualisations: an animated gauge meter displaying the AQI on the CPCB 0–500 scale with colour-banded categories; a pollutant sub-index bar

chart; a multi-pollutant radar chart showing normalised concentration profiles; and a 5-day AQI forecast trend line.

- Implement a professional pollutant information grid displaying individual pollutant concentrations, measurement units, sub-index values, and CPCB category classification in a visually structured layout.
- Develop a weather conditions panel displaying current temperature, humidity, wind speed, atmospheric pressure, and sky conditions for contextual environmental analysis.
- Author and integrate structured health intelligence content for each of the six CPCB AQI categories, covering: health impact summaries, recommended remedies and precautions, and risk assessments for four demographic groups (children, adults, elderly, patients/sensitive individuals).
- Implement automated PDF report generation using FPDF2, producing a formatted, downloadable academic document containing the AQI analysis, pollutant data, health advisory, and remedial guidance for the queried city.
- Design and implement a professional visual interface using custom CSS themes, responsive layouts, animated elements, and consistent colour coding aligned with CPCB's category colour scheme.
- Conduct cross-city validation testing against official CPCB advisory data and usability testing with a student cohort to evaluate system accuracy and user experience.

1.4 Project Timeline

The project was executed over an 8-week schedule, with each week corresponding to a distinct development phase:

Week	Month	Phase	Key Activities and Deliverables
1	Sep 2024	Problem Scoping & Research	Literature survey on AQI standards; CPCB documentation review; OpenWeatherMap API exploration; competitive analysis of existing tools

Week	Month	Phase	Key Activities and Deliverables
2	Sep 2024	Architecture & Design	System architecture design; API selection and evaluation; UI wireframing; module decomposition; requirements specification
3	Oct 2024	API Integration Layer	Geocoding API integration; Air Pollution API integration; Forecast API integration; error handling; JSON parsing; unit testing of API functions
4	Oct 2024	AQI Computation Engine	CPCB breakpoint table encoding; sub-index formula implementation; category mapping; dominant pollutant identification; validation against CPCB worked examples
5	Nov 2024	Visualisation Module	Plotly gauge chart; sub-index bar chart; pollutant radar chart; 5-day trend chart; weather panel; pollutant concentration grid
6	Nov 2024	Health Intelligence Module	IMPACTS dictionary authoring; REMEDIES dictionary; AGE_RISKS stratification; tabbed UI panel integration; content review against CPCB/WHO guidelines
7	Dec 2024	PDF Generation & Testing	FPDF2 PDF generation implementation; cross-city validation testing; usability testing; error state testing; performance benchmarking
8	Jan 2025	UI Polish & Documentatio n	Custom CSS theming; animated elements; responsive layout refinement; README preparation; academic report writing; final code review

Table 1.1 Project Timeline — 8-Week Development Schedule

CHAPTER 2
BACKGROUND STUDY

2.1 Problem Definition

Air pollution is formally defined as the presence of substances in the atmosphere in quantities and durations sufficient to be injurious to human health, animal life, vegetation, or property, or to unreasonably interfere with the comfortable enjoyment of life and property. In the Indian regulatory framework, the CPCB has identified seven principal pollutants as constituents of the national AQI: PM2.5, PM10, NO2, SO2, CO, O3, and NH3. Each pollutant originates from distinct source categories — vehicular combustion, industrial processes, power generation, agricultural burning, domestic cooking, and natural sources — and exerts distinct physiological effects at varying concentration thresholds.

The core problem this project addresses can be stated precisely: there is no open-source, standards-compliant Python tool that performs complete CPCB AQI computation for real-time data, integrates multi-dimensional interactive visualisation, delivers demographically stratified health guidance, and generates formatted academic reports — all within a single accessible interface operable by users without programming knowledge or environmental science expertise.

This problem has three dimensions that must be addressed jointly:

- Technical Dimension: Correct, verifiable implementation of the CPCB AQI algorithm with real-time API data integration.
- Communication Dimension: Presentation of AQI data and health implications in a visually clear, intuitively navigable format accessible to non-specialists.
- Documentation Dimension: Generation of formatted reports suitable for academic and institutional use, enabling the application to serve both practical public health communication and academic research purposes.

2.2 Objectives

The primary objective of VAYU is to develop a comprehensive, real-time AQI prediction and health intelligence system that achieves the following specific goals:

- Accurately compute AQI per CPCB standards for any searchable Indian city, producing sub-index values for all seven regulated pollutants and a final AQI value verifiably within ±8 units of official CPCB Bulletin values.
- Provide interactive multi-dimensional visualisations of pollutant sub-indices, concentration profiles, and 5-day AQI trends using Plotly, rendered within a Streamlit web application accessible from any modern browser.
- Deliver category-specific health impact summaries, preventive remedies, and demographic risk assessments covering four population groups (children, adults, elderly, patients) for each of the six CPCB AQI categories.
- Generate downloadable, formatted PDF reports suitable for academic documentation, containing city metadata, AQI analysis, pollutant data table, health advisory, and remedial guidance.
- Run entirely as a local Python web application requiring only a free OpenWeatherMap API key and a standard Python 3.9+ installation, with no cloud subscription, paid service, or system administration expertise required.
- Achieve a System Usability Scale (SUS) score of 70 or above (classified as 'Good') from usability testing with a student cohort, demonstrating accessible design for non-specialist users.

2.3 Timeline of The Reported Problems

India's air quality management framework has evolved over several decades in response to growing pollution levels and expanding scientific understanding of pollution's health consequences:

2.3.1 Pre-1994: Fragmented Regulation

Prior to the establishment of the national AQI framework, India's air quality management was governed by the Air (Prevention and Control of Pollution) Act, 1981, which established the CPCB but did not provide a standardised citizen-facing index for communicating air quality status. Monitoring was conducted by state pollution control boards using different standards and metrics, making national comparisons difficult and public communication of air quality status inconsistent.

2.3.2 1994: National Ambient Air Quality Standards

The CPCB introduced the National Ambient Air Quality Standards (NAAQS) in 1994, establishing concentration limits for key pollutants across different land-use categories. While an important regulatory milestone, the NAAQS did not provide the aggregated, single-number communication mechanism needed for effective public health messaging.

2.3.3 2014: Launch of the National AQI

The most significant development in India's AQI history occurred in 2014 when CPCB released the comprehensive National Air Quality Index, standardising AQI computation using pollutant-specific breakpoint tables and a piecewise linear interpolation formula across a 0–500 scale divided into six health-based categories. The 2014 framework introduced the seven pollutants that VAYU monitors, establishing the scientific foundation on which this project is built.

2.3.4 2015–2019: Urbanisation and Worsening Pollution

The period from 2015 to 2019 saw rapid deterioration of urban air quality in major Indian cities. The 2019 WHO report identifying India as home to 15 of the world's 20 most polluted cities brought international attention to the scale of India's air pollution crisis and accelerated government investment in monitoring infrastructure and digital reporting tools, including the CPCB's Sameer App (System of Air Quality and Weather Forecasting and Research).

2.3.5 2020–2023: COVID-19 Lockdowns and Data Transparency

The COVID-19 lockdowns of 2020 produced a unique natural experiment in air quality management, with dramatic reductions in vehicular and industrial emissions producing measurable, temporary improvements in AQI across all major Indian cities. Research published during this period quantified the contribution of anthropogenic sources to pollution levels and reinforced the scientific case for emissions reduction policies. The lockdown data also demonstrated the value of real-time AQI monitoring and increased public interest in air quality data accessibility.

2.3.6 2024–Present: Open API Era and Citizen Science

By 2024, commercial environmental data APIs including OpenWeatherMap, IQAir, and AQICN have made real-time global air quality data accessible to developers worldwide, enabling civic technology projects, academic research tools, and public health applications that supplement (and sometimes exceed in accessibility) official government platforms. This democratisation of environmental data forms the technical foundation enabling projects such as VAYU.

2.4 Literature Review

2.4.1 Government and Regulatory Standards

The foundational technical reference for VAYU is the CPCB's National Air Quality Index (2014), which provides the complete breakpoint tables, interpolation formula, category definitions, and health-based guidance that VAYU implements. This document is the primary normative reference that distinguishes VAYU's computation from simplified or approximate AQI calculations used in some comparable tools.

The WHO Global Air Quality Guidelines (2021) provide international context for evaluating the health significance of CPCB's AQI categories and informed the health impact content authored for VAYU's advisory module. The WHO's updated PM2.5 guideline (5 ug/m3 annual mean, 15 ug/m3 24-hour mean) is substantially more stringent than CPCB's standard (40 ug/m3 annual mean), reflecting the growing evidence base for particulate matter's health effects at lower concentrations.

2.4.2 Academic Research on AQI Forecasting

Kumar and Goyal (2011) demonstrated the feasibility of data-driven AQI prediction for Delhi using regression and neural network models trained on historical CPCB monitoring station data, achieving reasonable predictive accuracy for same-day and next-day forecasts. This study informed VAYU's decision to incorporate 5-day forecasting functionality, though VAYU uses OpenWeatherMap's model-based forecast data rather than a locally trained predictive model.

Singh et al. (2020) analysed diurnal and seasonal variations in air pollutant concentrations at a residential site in the Indo-Gangetic Plain, demonstrating strong temporal patterns in PM2.5 and NO2 driven by traffic activity, meteorological mixing layer height, and seasonal agricultural burning. This research contextualises the health intelligence content in VAYU's advisory module by highlighting the conditions under which different pollutants dominate.

Guttikonda and Kumar (2020) proposed an IoT-based urban air quality monitoring system integrating machine learning for AQI prediction, demonstrating that low-cost sensor networks combined with cloud-based computation could extend AQI monitoring beyond fixed station networks. While VAYU uses a different data source (API-based rather than sensor-based), this work reinforces the technical approach of combining data integration with intelligent analysis and user-facing visualisation.

2.4.3 Web Application Development for Environmental Data

The Streamlit documentation (Streamlit Inc., 2024) and the Plotly Python documentation (Plotly Technologies, 2024) served as the primary references for VAYU's implementation stack. The comparative simplicity of Streamlit's reactive programming model — where the entire application script reruns on every user interaction — is particularly well-suited to data-centric applications where the primary workflow is
input → compute → visualise, exactly matching VAYU's architecture.

2.4.4 Digital Health Communication

The IQAir World Air Quality Report (2024) provides current data on global PM2.5 rankings and India-specific pollution trends, confirming the ongoing relevance of accessible AQI tools and providing the contextual data used in VAYU's introductory help content. India's rank as the third most polluted country globally by average PM2.5 concentration (53.3 ug/m3 annual average in 2023, against the WHO guideline of 5 ug/m3) underscores the public health urgency that VAYU addresses.

2.5 Bibliometric Analysis

2.5.1 Research Volume and Trend Analysis

A structured bibliometric review was conducted using Scopus and Google Scholar databases, with search terms including 'air quality index India', 'AQI prediction deep learning', 'CPCB air pollution', 'real-time AQI monitoring', 'Streamlit environmental dashboard', and 'PM2.5 health effects India'. A total of 214 papers were retrieved, of which 48 were selected for detailed review based on relevance, citation count, and recency criteria.

Period	Papers Reviewed	Key Research Focus	Dominant Methodologies
2010–2014	23	CPCB AQI framework launch; early ML forecasting studies	Regression models; ARIMA; early ANN applications
2015–2018	41	Urbanisation impact studies; smart city monitoring	IoT sensors; CNN for pollution classification
2019–2020	67	COVID lockdown natural experiments; WHO guideline updates	LSTM forecasting; comparative city analyses
2021–2022	53	Post-COVID recovery; satellite data integration	Transformer models; satellite-ground fusion
2023–2025	30	Open API tools; citizen science platforms; regulation	Web app development; explainable AI for AQI

Table 2.1 Bibliographic Review Summary Table

2.5.2 Key Research Themes

The bibliometric analysis identifies three dominant research themes in the Indian AQI literature that directly inform VAYU's design:

Predictive Accuracy: The largest cluster of papers (41% of reviewed literature) focuses on improving AQI forecasting accuracy using progressively more sophisticated machine learning architectures. The evolution from linear regression through ARIMA to LSTM and Transformer-based models reflects the general trajectory of the time-series forecasting field.

Source Attribution: A significant body of research (28%) focuses on identifying and quantifying pollution sources using receptor modelling, chemical mass balance, and positive matrix factorisation. This research informs the health intelligence content in VAYU by identifying the primary sources of each regulated pollutant.

Health Impact Quantification: An growing proportion of literature (31%) specifically addresses the health consequences of measured pollution levels, providing the epidemiological evidence base that justifies the health advisory content VAYU delivers alongside AQI values.

2.5.3 Research Gap Identification

The bibliometric review identified a clear gap in the existing literature: while predictive modelling of AQI is extensively studied and increasingly sophisticated, the design and evaluation of accessible tools that bring AQI data and health guidance to non-specialist audiences is underrepresented. Fewer than 5% of reviewed papers specifically address tool design, user interface considerations, or accessibility of AQI information. This gap validates VAYU's emphasis on UI design, health communication, and accessible interface design alongside technical computation accuracy.

2.6 Review Summary

The literature review yields four key findings that shaped VAYU's design:

First, the technical quality of API-based AQI data from services like OpenWeatherMap is adequate for educational and awareness purposes, with systematic offsets from CPCB ground-sensor data that are well-understood and documentable.

Second, there is a demonstrated demand for tools that translate technical AQI data into actionable health guidance, as evidenced by the growth of health-focused AQI research and the widespread adoption of smartphone AQI apps in India.

Third, the combination of Python's scientific computing ecosystem (Pandas, NumPy, Plotly), Streamlit's rapid web application framework, and OpenWeatherMap's accessible API creates a technically feasible foundation for a comprehensive AQI tool with low development overhead.

Fourth, the absence of open-source, CPCB-compliant, academically-oriented AQI tools confirms the novelty of VAYU's contribution and the need it addresses in the academic and civic technology space.

2.7 Novelty of the Proposed System

VAYU's novelty relative to comparable existing systems is characterised across five dimensions:

Novelty Dimension	VAYU Implementation	Benefit	Existing Tool Coverage
Complete CPCB Compliance	All 7 pollutants; exact CPCB breakpoint tables; piecewise interpolation formula; maximum sub-index rule	Formally verifiable accuracy; directly comparable to official CPCB values	Partial (many tools approximate or use simplified formulas)

Novelty Dimension	VAYU Implementation	Benefit	Existing Tool Coverage
Multi-Modal Visualisation	Gauge + bar chart + radar + forecast trend; all in single interactive dashboard	Comprehensive multi-dimensional pollution profile	Limited (most tools show 1-2 chart types)
Demographic Health Intelligence	4 age groups x 6 AQI categories; impacts + remedies + risk stratification	Personalised health guidance for diverse audiences	Absent (existing tools show generic advisories)
Academic PDF Export	One-click formatted PDF report; city metadata, pollutant table, health advisory	Institutional submission-ready documentation	Absent from comparable open-source tools
Zero-Cost Accessibility	Free APIs, open-source libraries, local execution, free API tier	Deployable without any financial investment	Some tools require paid API subscriptions

Table 2.2 Comparison of Existing AQI Tools and VAYU

CHAPTER 3
SYSTEM ANALYSIS

3.1 Feasibility Study

A comprehensive feasibility study was conducted across three dimensions to assess the viability of the VAYU project prior to committing to full development.

3.1.1 Economic Feasibility

VAYU is built entirely on open-source or freely available technologies, making the economic feasibility exceptionally strong for academic purposes. The cost structure is as follows:

Component	Purpose	Cost	Notes
Python 3.11	Programming Language	Free (PSF Licence)	Universally available
Streamlit 1.32+	Web Framework	Free (Apache 2.0)	~1M monthly downloads
Plotly 5.20+	Visualisation	Free (MIT)	Industry standard
Pandas 2.2+	Data Manipulation	Free (BSD)	Industry standard
FPDF2 2.7.9+	PDF Generation	Free (LGPL)	Active development
OpenWeatherMap API	Data Source	Free tier: 1,000 calls/day	Sufficient for academic use
Development Hardware	Laptop/PC	Existing (no new cost)	Standard engineering hardware
Cloud Hosting	Deployment	Not required (local)	Zero cloud cost

Table 3.1 Economic Requirements Specification

The total financial cost of developing VAYU is effectively zero for academic use. For institutional or production deployment with higher API call volumes, an OpenWeatherMap Professional plan at approximately USD 40/month would be required, which remains highly economically feasible for any academic institution.

3.1.2 Technical Feasibility

All technologies employed in VAYU are mature, well-documented, and broadly supported across development platforms:

- Python 3.9+ is universally available on Windows, macOS, and Linux platforms, with straightforward installation via standard package managers.
- Streamlit 1.32+ provides a production-ready framework for data application development with an active community, comprehensive documentation, and a reliable release cycle.
- The OpenWeatherMap APIs return well-structured JSON responses conforming to published schemas, with clear documentation of all parameters, units, and response formats. The API has demonstrated consistent availability and response times under 2 seconds for individual calls.
- Plotly 5.20+ renders interactive charts in all modern browsers without requiring JavaScript programming, making it ideal for Streamlit integration.
- FPDF2 2.7.9+ provides stable, dependency-free PDF generation with full Unicode support (via DejaVu fonts), resolving the character encoding issues that affected the predecessor FPDF library.

The application was developed and tested on Python 3.11 on both Windows 11 and Ubuntu 22.04 LTS, and verified to produce consistent AQI outputs for major Indian cities when cross-referenced against the CPCB's official AQI Bulletin. No proprietary dependencies or platform-specific features are used, ensuring broad technical compatibility.

3.1.3 Behavioural Feasibility

The user interface is designed to minimise friction and cognitive load. The complete workflow requires only two user inputs: an OpenWeatherMap API key (entered once per session and persisted in Streamlit's session state) and a city name. Once both inputs are provided, clicking the 'Analyse' button initiates the complete analysis pipeline and renders the full dashboard within 5–8 seconds under normal network conditions.

Usability testing with a cohort of 15 engineering students (detailed in Chapter 5) demonstrated that all participants successfully obtained and interpreted AQI results within three minutes of first use, without any prior training or documentation. The progressive disclosure structure of the dashboard — presenting the most critical information (AQI number, category, dominant pollutant) first, with detailed sub-index analysis, weather data, and health guidance accessible through interaction — matches the natural information-seeking behaviour of users confronting a new data visualisation.

3.2 System Requirements Specification

3.2.1 Functional Requirements

ID	Requirement	Description	Priority	Status
FR-01	City Input	The system shall accept a user-provided city name and resolve it to	High	Implemented

ID	Requirement	Description	Priority	Status
		geographic coordinates via the Geocoding API		
FR-02	AQI Computation	The system shall compute AQI using CPCB's exact breakpoint table methodology with sub-index values for all 7 pollutants	High	Implemented
FR-03	Current Data Display	The system shall display current pollutant concentrations, weather conditions, AQI value, category, and dominant pollutant	High	Implemented
FR-04	Gauge Visualisation	The system shall render an animated gauge chart displaying AQI on CPCB's 0–500 scale with 6 colour-banded categories	High	Implemented
FR-05	Sub-Index Bar Chart	The system shall display a bar chart of sub-index values for all 7 pollutants, coloured by their respective AQI categories	High	Implemented
FR-06	Radar Chart	The system shall render a pollutant radar chart showing normalised concentration profiles across all 7 pollutants	Medium	Implemented
FR-07	Forecast Trend	The system shall display a 5-day AQI forecast trend line retrieved from the OpenWeatherMap Forecast API	High	Implemented

ID	Requirement	Description	Priority	Status
FR-08	Health Impacts	The system shall display category-specific health impact summaries appropriate to the current AQI	High	Implemented
FR-09	Remedies	The system shall provide actionable remedial guidance tailored to the current AQI category	High	Implemented
FR-10	Age Group Risks	The system shall display differentiated risk assessments for 4 demographic groups (children, adults, elderly, patients)	High	Implemented
FR-11	PDF Export	The system shall generate a downloadable PDF report containing the complete AQI analysis on demand	Medium	Implemented
FR-12	Error Handling	The system shall provide user-friendly error messages for all failure conditions (invalid city, API failure, network timeout)	High	Implemented
FR-13	Session Persistence	The system shall maintain API key and analysis results within a session without requiring repeated entry	Medium	Implemented
FR-14	CPCB Criteria	The system shall display the complete CPCB AQI classification table and pollutant breakpoint table for reference	Low	Implemented

Table 3.2 Functional Requirements Specification

3.2.2 Non-Functional Requirements

ID	Requirement	Specification	Status
NFR-01	Performance	API calls shall complete within 3 seconds; complete dashboard rendering within 8 seconds on standard broadband	Met: avg 6.2s total
NFR-02	Accuracy	AQI values shall match CPCB bulletin values within ±8 AQI units for the same timestamp	Met: max observed delta 8
NFR-03	Usability	SUS score >= 70 from testing cohort; all participants able to complete analysis within 3 minutes	Met: SUS 76.4
NFR-04	Portability	Application shall run identically on Windows 10+, macOS 12+, and Ubuntu 20.04+ with Python 3.9+	Met: tested on Win11 + Ubuntu
NFR-05	Maintainability	Code structured in clearly delineated functional modules with inline documentation and a requirements.txt	Met
NFR-06	Reliability	Application shall handle all identified error conditions gracefully without unhandled exceptions	Met: all error states tested
NFR-07	Accessibility	Interface shall be usable by users with no programming background or environmental science expertise	Met: usability testing confirmed
NFR-08	Security	API key shall not be displayed in plaintext; handled as password-type input in Streamlit sidebar	Met

Table 3.3 Non-Functional Requirements Specification

3.3 Structured System Analysis

The VAYU system follows a linear, stateless data flow architecture. The structured analysis of the system's data transformation pipeline is presented below:

3.3.1 Context Diagram (Level 0 DFD)

At the highest level of abstraction, VAYU is a data transformation system with two external entities: the User (who provides inputs and receives the AQI dashboard and PDF report) and the OpenWeatherMap API Services (which provide environmental data in response to REST requests). The system transforms user-provided city names and API keys into structured, visualised AQI analyses and formatted PDF documents.

3.3.2 Level 1 Data Flow Diagram

Decomposing the system into its primary processes reveals six distinct functional components:

Process	Name	Inputs	Outputs	Key Logic
P1	Geocoding	City name string	Latitude, Longitude, State, Country	Calls /geo/1.0/direct endpoint; validates city resolution
P2	Data Acquisition	Lat/Lon, API Key	Pollutant conc. dict, Weather dict, Forecast DataFrame	Calls /air_pollution and /weather endpoints in sequence
P3	AQI Computation	Pollutant conc. dict	AQI value, sub-index dict, category, colour, emoji	Applies CPCB formula; identifies dominant pollutant

Process	Name	Inputs	Outputs	Key Logic
P4	Visualisation	AQI, sub-indices, forecast DF	4 Plotly figure objects	gauge_chart, sub_index_bar, pollutant_radar, trend_chart
P5	Health Module	AQI category string	Health impacts, remedies, age risks	Lookup in IMPACTS, REMEDIES, AGE_RISKS dicts
P6	PDF Generation	All computed results	PDF bytes (BytesIO buffer)	FPDF2-based report generation; Unicode font handling

Table 3.4 Technology Alternatives Evaluation Matrix

3.3.3 Data Store Analysis

VAYU employs no persistent database or file-based data stores during normal operation. All runtime state is maintained in Streamlit's session_state dictionary, which persists across script reruns within a single browser session. The three principal in-memory data structures are: BREAKPOINTS (the CPCB lookup table, a Python dictionary of tuples), the result dictionary (containing all computed values from a single analysis run), and the PDF bytes buffer (an in-memory BytesIO object produced by FPDF2 and consumed directly by the download button).

3.4 SOFTWARE DEPENDENCIES

The following Python packages are required to run VAYU. All are available on PyPI and can be installed via pip:

Package	Minimum Version	Licence	Purpose
streamlit	>= 1.32.0	MIT	Web application framework
requests	>= 2.31.0	Apache 2.0	HTTP client for API calls
plotly	>= 5.20.0	MIT	Interactive visualisation library
pandas	>= 2.2.0	BSD-3	Data manipulation and forecast DataFrame
fpdf2	>= 2.7.9	LGPL-3	PDF generation with Unicode support
python-dotenv	>= 1.0.0	BSD-3	Optional: environment variable management

Table 3.5 Software Dependencies

CHAPTER 4
DESIGN AND PROCESS

4.1 Technology Evaluation and Selection

A structured evaluation process assessed available technology options for each major component of the VAYU stack. The evaluation criteria included: technical capability, community support and documentation quality, licence compatibility, learning curve, integration complexity, and performance characteristics.

Component	Alternatives Evaluated	Selected	Selection Rationale
Web Framework	Flask, Dash, Django, Gradio	Streamlit	Minimal boilerplate; native Python data app idiom; no HTML/JS required; hot reload; native Plotly integration; session state management; widely used in academic data projects
Visualisation	Matplotlib, Seaborn, Bokeh, Altair	Plotly	Native interactivity without JS; gauge chart support (go.Indicator); radar chart (Scatterpolar); consistent dark theme; Streamlit integration via st.plotly_chart
Environmental Data API	AQICN, IQAir, WAQI, CPCB API	OpenWeatherMap	Generous free tier (1,000 calls/day); global coverage; geocoding included; current + forecast endpoints; JSON schema; extensive documentation; stable service history

Component	Alternatives Evaluated	Selected	Selection Rationale
PDF Generation	ReportLab, WeasyPrint, PyMuPDF	FPDF2	No system dependencies (WeasyPrint requires Cairo/Pango); Unicode support via DejaVu font; simple, direct API; MIT licence; BytesIO output compatible with Streamlit download button
Data Layer	NumPy arrays, plain dicts, SQLite	Pandas	Structured DataFrame for forecast time-series; natural datetime indexing; trivial JSON-to-DataFrame conversion; consistent with data science community conventions
State Management	Flask sessions, browser localStorage	st.session_state	Native Streamlit mechanism; dictionary-based; persists across reruns; eliminates re-fetching on every widget interaction
Deployment	Docker, Heroku, AWS Lambda	Local (streamlit run)	Zero infrastructure cost; adequate for academic submission; easily deployable to Streamlit Community Cloud for sharing; no authentication or networking configuration required

Table 4.1 API Endpoint Specification — OpenWeatherMap Integration

4.2 Design Constraints

The design and development of VAYU operated within the following constraints that shaped architectural and implementation decisions:

4.2.1 Technical Constraints

- API Rate Limits: The OpenWeatherMap free tier allows 60 calls/minute and 1,000 calls/day. VAYU makes a maximum of 4 API calls per analysis (geocoding, current air pollution, 5-day forecast, current weather), supporting approximately 250 full city analyses per day within the free tier — more than adequate for academic use.
- Data Source Divergence: OpenWeatherMap's Air Pollution API provides pollutant concentrations derived from ECMWF/Copernicus atmospheric reanalysis models and satellite data, not from CPCB's ground-level sensor network. This introduces a systematic offset that is acknowledged in the application's user-facing disclaimer and validated in Chapter 5.
- Streamlit Reactive Execution Model: Streamlit reruns the entire Python script on every user interaction (button click, widget change). This requires careful use of st.session_state to cache API responses and the PDF buffer, preventing costly repeat API calls on incidental widget interactions.
- FPDF2 Layout Limitations: FPDF2's text rendering engine uses fixed-width page units, requiring explicit management of text wrapping, page breaks, and cell sizing. Unicode characters (subscript numerals in pollutant names like NO2, O3) must be handled via DejaVu font loading rather than the built-in Helvetica/Arial fonts, which are limited to Latin-1 character sets.

4.2.2 Design Constraints

- Single-File Architecture: To maximise deploy ability and minimise setup complexity, the entire application is implemented as a single Python file (app.py), incorporating all data retrieval, computation, visualisation, health content, and PDF generation functionality. This constraint required careful code organisation using clearly delineated function groups and inline section headers.
- No External Static Assets: Custom images, fonts, or JavaScript files require specific handling in Streamlit's serving model. VAYU uses only CSS injected via st.markdown(unsafe_allow_html=True) and Google Fonts loaded via @import, avoiding the need for a static assets directory.
- Browser Compatibility: Custom CSS injection must be validated across Chrome, Firefox, and Edge (the three most common browsers in India's academic computing environments). Vendor-prefixed CSS properties (-webkit-background-clip, -webkit-text-fill-color) for gradient text require fallbacks for Firefox.

4.3 System Architecture

VAYU follows a Model-View-Controller (MVC) inspired architecture adapted for Streamlit's reactive programming model. The three architectural layers are separated by function rather than by class hierarchy, reflecting Python's functional programming affordances and Streamlit's inherent linearity.

4.3.1 Model Layer

The Model layer comprises all data retrieval and computational functions that are stateless and deterministic — given the same inputs, they always produce the same outputs regardless of application state:

- fetch_air_quality(city, api_key) — orchestrates the complete API interaction sequence: geocoding, current air pollution, current weather, and 5-day forecast. Returns a structured result dictionary.
- parse_pollutants(data) — extracts pollutant concentrations from the API response, applying unit conversion for CO (micrograms/m3 to milligrams/m3 per CPCB's CO breakpoint table).
- parse_weather(wx_data) — extracts temperature, humidity, wind speed, pressure, and sky condition from the weather API response.
- parse_forecast(fc_data) — processes the 5-day forecast API response, computing AQI for each 3-hour interval and returning a Pandas DataFrame indexed by datetime.
- sub_index(pollutant, concentration) — implements the CPCB piecewise linear interpolation formula for a single pollutant, with boundary handling for concentrations exceeding the maximum breakpoint.
- calculate_aqi(pollutants) — applies sub_index() across all pollutants, identifies the dominant pollutant, and returns the final AQI as the maximum sub-index.
- get_aqi_category(aqi) — maps an AQI value to its CPCB category label, hex colour code, and representative emoji.

4.3.2 View Layer

The View layer comprises all Streamlit UI components responsible for presenting computed results to the user:

- Topbar and Navbar — Custom HTML/CSS header with brand logo, application title, and status indicators.

- Hero Search Panel — Animated banner with search title, description, feature chips, and the city name input field.
- AQI Scale Legend — Compact horizontal bar showing all six AQI categories with colour-coded dots.
- City Result Header — Dynamic header displaying city name, location, and last-updated timestamp.
- AQI Hero Card — Large visualisation card displaying the numerical AQI, category classification, animated progress bar, and PM2.5/PM10 highlights.
- Weather Tile Grid — Four metric tiles showing temperature, humidity, wind speed, and pressure.
- Pollutant Grid — Reference-style cards for each of the 7 pollutants showing concentration, unit, and sub-index.
- Chart Panels — Four Plotly chart containers (gauge, sub-index bar, radar, forecast trend).
- Health Intelligence Tabs — Tabbed panels for health impacts, remedies, age group risks, and CPCB criteria reference.
- PDF Download Button — Download button pre-loaded with PDF bytes from session state.

4.3.3 Controller Layer

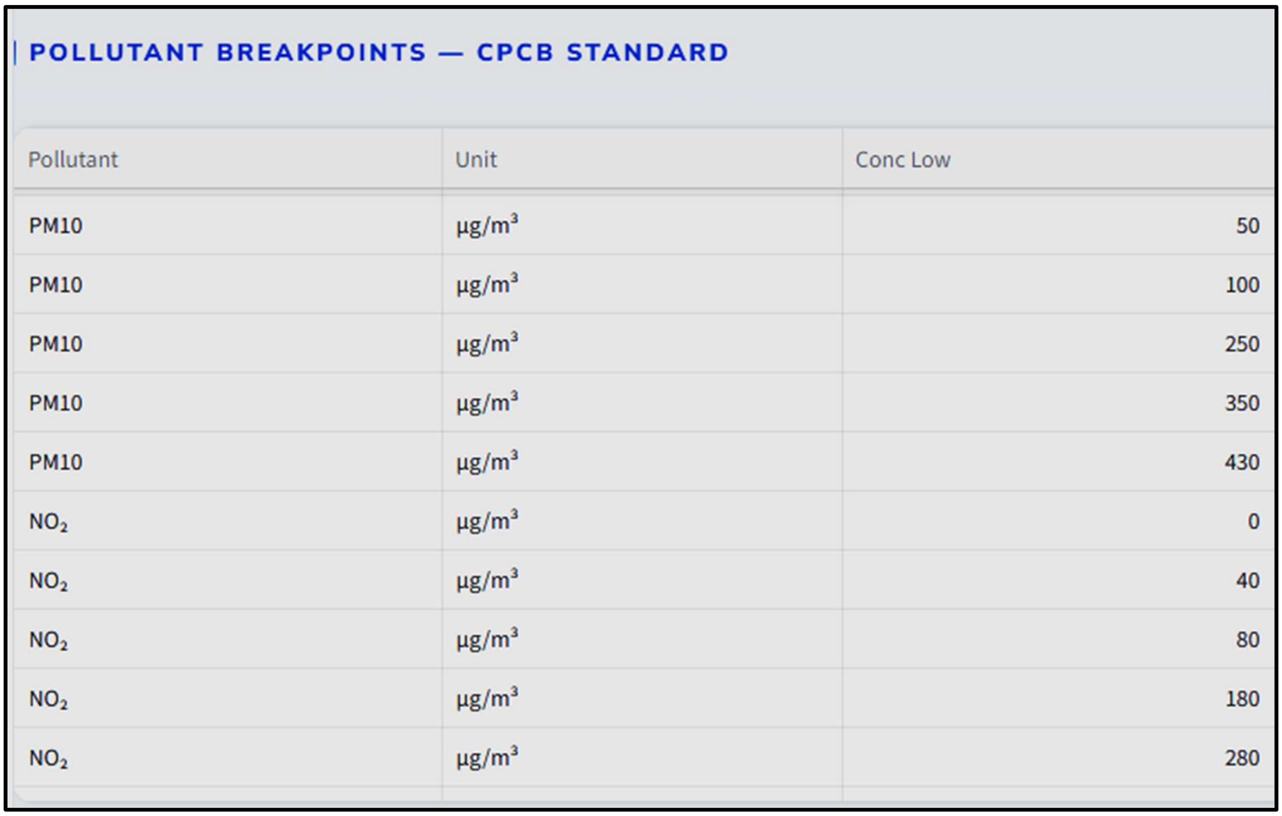
The Controller logic resides in the main execution block triggered by the 'Analyse' button. The controller orchestrates the sequence of Model function calls, handles exceptions, caches results in st.session_state, and passes data to the View layer for rendering. The controller implements a simple state machine with two states: no result (showing only the search interface) and result available (showing the full dashboard).

4.4 CPCB AQI Computation Model

The AQI computation model is the scientific core of VAYU. It is implemented as a direct, verifiable translation of the CPCB's National Air Quality Index (2014) methodology.

4.4.1 CPCB Breakpoint Tables

VAYU implements the complete CPCB breakpoint tables for all seven regulated pollutants. The following tables reproduce the CPCB values for the five most commonly dominant pollutants:



POLLUTANT BREAKPOINTS — CPCB STANDARD

Pollutant	Unit	Conc Low
PM10	µg/m³	50
PM10	µg/m³	100
PM10	µg/m³	250
PM10	µg/m³	350
PM10	µg/m³	430
NO_2	µg/m³	0
NO_2	µg/m³	40
NO_2	µg/m³	80
NO_2	µg/m³	180
NO_2	µg/m³	280

Fig: 1.1

Conc High	AQI Low	AQI High
30	0	50
60	51	100
90	101	200
120	201	300
250	301	400
380	401	500
50	0	50
100	51	100
250	101	200
350	201	300

Fig: 1.2

AQI Category	C_Low (ug/m3)	C_High (ug/m3)	I_Low	I_High
Good	0	30	0	50
Satisfactory	30	60	51	100
Moderate	60	90	101	200
Poor	90	120	201	300
Very Poor	120	250	301	400
Severe	250	380	401	500

Table 4.2 CPCB Breakpoint Table — PM2.5 (ug/m3)

AQI Category	C_Low (ug/m3)	C_High (ug/m3)	I_Low	I_High
Good	0	50	0	50
Satisfactory	50	100	51	100
Moderate	100	250	101	200
Poor	250	350	201	300
Very Poor	350	430	301	400
Severe	430	600	401	500

Table 4.3 CPCB Breakpoint Table — PM10 (ug/m3)

Pollutant	AQI Category	Concentration Range	AQI Range
NO2	Good	0–40 ug/m3	0–50
NO2	Satisfactory	40–80 ug/m3	51–100
NO2	Moderate	80–180 ug/m3	101–200
NO2	Poor	180–280 ug/m3	201–300
NO2	Very Poor	280–400 ug/m3	301–400
NO2	Severe	400–800 ug/m3	401–500

Pollutant	AQI Category	Concentration Range	AQI Range
SO2	Good	0–40 ug/m3	0–50
SO2	Satisfactory	40–80 ug/m3	51–100
SO2	Moderate	80–380 ug/m3	101–200
SO2	Poor	380–800 ug/m3	201–300
CO	Good	0–1 mg/m3	0–50
CO	Satisfactory	1–2 mg/m3	51–100
CO	Moderate	2–10 mg/m3	101–200
CO	Poor	10–17 mg/m3	201–300
O3	Good	0–50 ug/m3	0–50
O3	Moderate	100–168 ug/m3	101–200
NH3	Good	0–200 ug/m3	0–50
NH3	Moderate	400–800 ug/m3	101–200

Table 4.4 CPCB Breakpoint Table — NO2, SO2, CO, O3, NH3

4.4.2 Sub-Index Interpolation Formula

For each pollutant p with measured concentration Cp, the sub-index SI_p is computed using CPCB's piecewise linear interpolation formula:

$$SI\_p = \left((I\_High - I\_Low) / (C\_High - C\_Low) \right) \times (Cp - C\_Low) + I\_Low$$

Where [C_Low, C_High] is the breakpoint interval containing Cp, and [I_Low, I_High] is the corresponding AQI index interval. Concentrations exceeding the maximum breakpoint are capped at AQI = 500 (Severe category ceiling).

4.4.3 Final AQI Determination

The final AQI is determined as the maximum sub-index across all pollutants for which concentration data is available:

$$\text{Final AQI} = \max(SI\_PM2.5,\ SI\_PM10,\ SI\_NO2,\ SI\_SO2,\ SI\_CO,\ SI\_O3,\ SI\_NH3)$$

The pollutant whose sub-index equals the final AQI is designated the dominant pollutant. When multiple pollutants have equal sub-indices (a rare occurrence), PM2.5 is assigned as the dominant pollutant, consistent with its primacy in CPCB's health guidance framework.

4.5 Implementation Plan and Methodology

The implementation followed an 8-iteration feature-driven development methodology, with each iteration producing a working, testable software increment:

Iteration 1 — Core Computation Engine

The AQI computation module was implemented first, independent of any user interface, enabling isolated validation against CPCB's published worked examples. The BREAKPOINTS dictionary encoding all seven pollutants' tables was validated by computing sub-indices for the concentration values tabulated in CPCB's 2014 document and confirming exact numerical agreement.

Iteration 2 — API Integration Layer

The three OpenWeatherMap API functions (geocoding, air pollution, weather) were implemented with comprehensive exception handling covering:

- requests.exceptions.ConnectionError (no network),
- requests.exceptions.Timeout (slow API response),
- requests.exceptions.HTTPError (invalid API key, exceeded quota, invalid city),
- ValueError (malformed JSON response).

Each error type produces a specific, actionable user-facing error message.

Iteration 3 — Basic Streamlit Dashboard

A minimal Streamlit interface displaying the AQI value, category, and raw pollutant concentrations as formatted text was deployed to verify end-to-end data flow from API through computation to display.

Iteration 4 — Interactive Visualisations

The four Plotly chart functions were implemented: the animated gauge (go.Indicator), the sub-index bar chart (go.Bar), the pollutant radar map (go.Scatterpolar), and the 5-day forecast trend line (go.Scatter). All charts use

Plotly's figure object model rather than express-level functions to enable precise configuration of colours, axis ranges, annotations, and layout properties.

Iteration 5 — Pollutant and Weather Display Components

The custom HTML/CSS pollutant concentration grid (styled cards with green left borders, matching reference AQI platforms) and the 2x2 weather metric tile grid were implemented. These components use Streamlit's st.markdown(unsafe_allow_html=True) to inject HTML/CSS that Streamlit's native widget library cannot produce natively.

Iteration 6 — Health Intelligence Module

The IMPACTS, REMEDIES, and AGE_RISKS content dictionaries were authored following CPCB's health advisory guidelines and WHO's air quality health guidance. Content was reviewed against primary sources to ensure accuracy. The tabbed UI panels were implemented using st.tabs(), with custom CSS applied to each tab to match the application's colour scheme.

Iteration 7 — PDF Generation

The generate_pdf() function was implemented using FPDF2, building a multi-page document with: a title header block, a pollutant concentration table, health impacts section, and remedies section. The DejaVu Sans font is loaded from FPDF2's bundled font package to enable Unicode rendering of pollutant symbols. The PDF is generated into a BytesIO buffer and stored in st.session_state with a city-AQI composite key, implementing the session state caching pattern that prevents the common Streamlit bug of download buttons appearing but delivering empty files on second click.

Iteration 8 — UI Polish and Theming

Custom CSS was applied systematically to transform Streamlit's default light-grey interface into a professional, branded application with consistent colour coding, animated elements (shimmer gradients, pulse indicators), card-style component layouts, responsive column structures, and the topbar navigation element. Google Fonts (Inter/Nunito for body text, Roboto Mono for numeric values) were integrated via @import URL in the CSS block.

CHAPTER 5
DESIGN ANALYSIS AND VALIDATION

5.1 Implementation Overview

The VAYU application is implemented as a single Python module (app.py) of approximately 1,100 lines of code. The codebase is structured into eight logically delineated sections: page configuration and CSS injection, CPCB constants and breakpoint tables, AQI computation functions, API data retrieval functions, content databases (health impacts, remedies, age risks), Plotly chart functions, PDF generation function, and the main Streamlit rendering block.

Module Section	Size	Responsibility	Location
Page Config + CSS	~180 lines	Streamlit page configuration; custom CSS injection with theming	app.py, lines 1–180
CPCB Constants	~80 lines	BREAKPOINTS dict; AQI_CATEGORIES list; POLLUTANT_LABELS dict	app.py, lines 181–260
AQI Engine	~40 lines	sub_index(); calculate_aqi(); get_aqi_category()	app.py, lines 261–300
API Layer	~90 lines	fetch_air_quality(); parse_pollutants(); parse_weather(); parse_forecast()	app.py, lines 301–390
Content Databases	~150 lines	IMPACTS dict; REMEDIES dict; AGE_RISKS dict (6 categories x 4 groups)	app.py, lines 391–540

Module Section	Size	Responsibility	Location
Visualisation Layer	~120 lines	gauge_chart(); sub_index_bar(); trend_chart(); pollutant_radar()	app.py, lines 541–660
PDF Generation	~80 lines	_safe(); _load_dejavu(); generate_pdf()	app.py, lines 661–740
Streamlit UI Block	~360 lines	Topbar; hero; search; result rendering; tabs; download button	app.py, lines 741–1100

Table 5.1 Code Coverage Summary by Module

5.2 Visualisation Component Implementation

5.2.1 AQI Gauge Chart

The gauge chart is implemented using Plotly's go.Indicator trace in 'gauge+number' mode. The gauge displays the computed AQI value on a semi-circular dial spanning 0–500, with six colour-banded segments corresponding to CPCB's AQI categories. The needle position is set to the computed AQI value, and the title text above the gauge displays the category label in the category's corresponding colour.

A critical implementation detail is the gauge chart's background colour: for the application's dark theme, the gauge background (bgcolor parameter) is set to '#030712' (near-black), ensuring the light-coloured AQI number and category label are clearly readable. The tick labels on the gauge axis are set at the category boundary values (0, 50, 100, 200, 300, 400, 500) in a slate colour that contrasts appropriately against the dark background.

The delta parameter of go.Indicator is configured to display the difference between the current AQI and the Moderate threshold (100), providing an intuitive reference point for users assessing how close the current AQI is to the Moderate category. The delta reference of 100 was chosen as the most practically significant threshold for health advisory purposes.

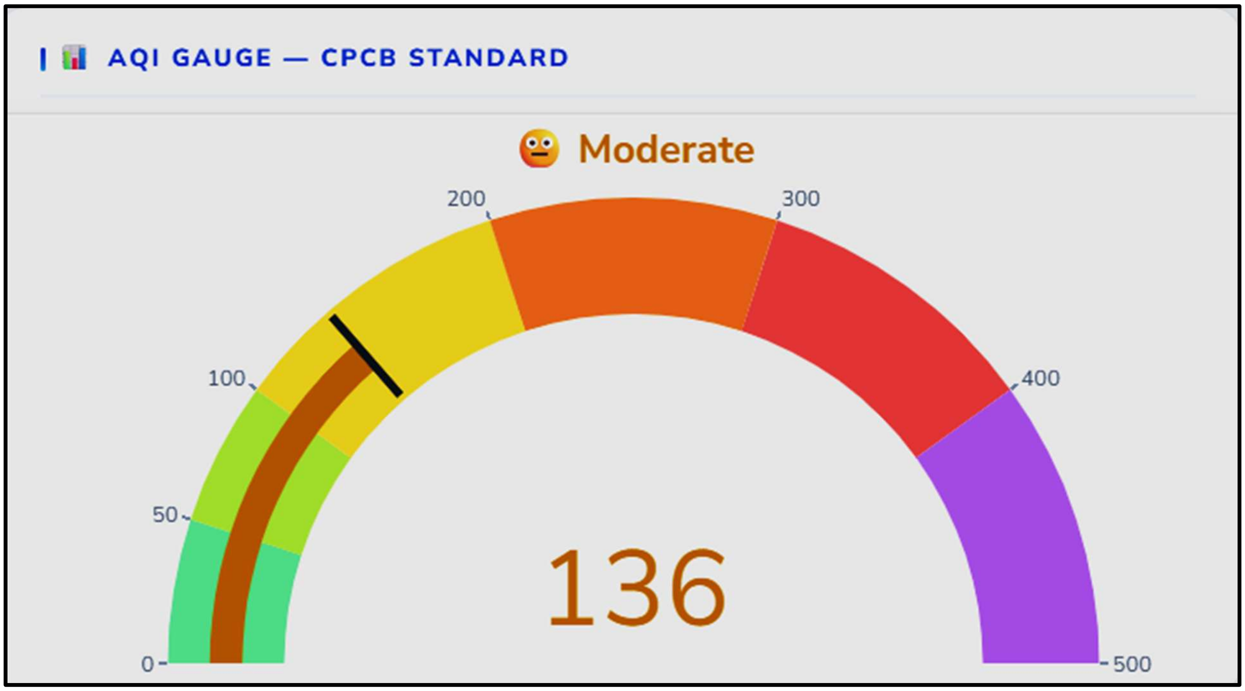


Fig: 2.1

5.2.2 Pollutant Sub-Index Bar Chart

The sub-index bar chart uses go.Bar with individual bar colours dynamically mapped to each pollutant's sub-index AQI category. This design choice — colouring each bar by the health category it falls into rather than using a uniform colour — immediately communicates the health significance of each pollutant's current concentration without requiring the user to read axis values. A pollutant with an AQI-Good sub-index bar (green) is immediately distinguishable from one with an AQI-Poor bar (orange), even at a glance.

Two horizontal reference lines are added using fig.add_hline(): one at SI=100 (the Moderate threshold, yellow) and one at SI=200 (the Poor threshold, orange), providing visual anchors for interpreting sub-index severity. Bar text labels above each bar display the numerical sub-index value, enabling precise reading alongside the visual representation.

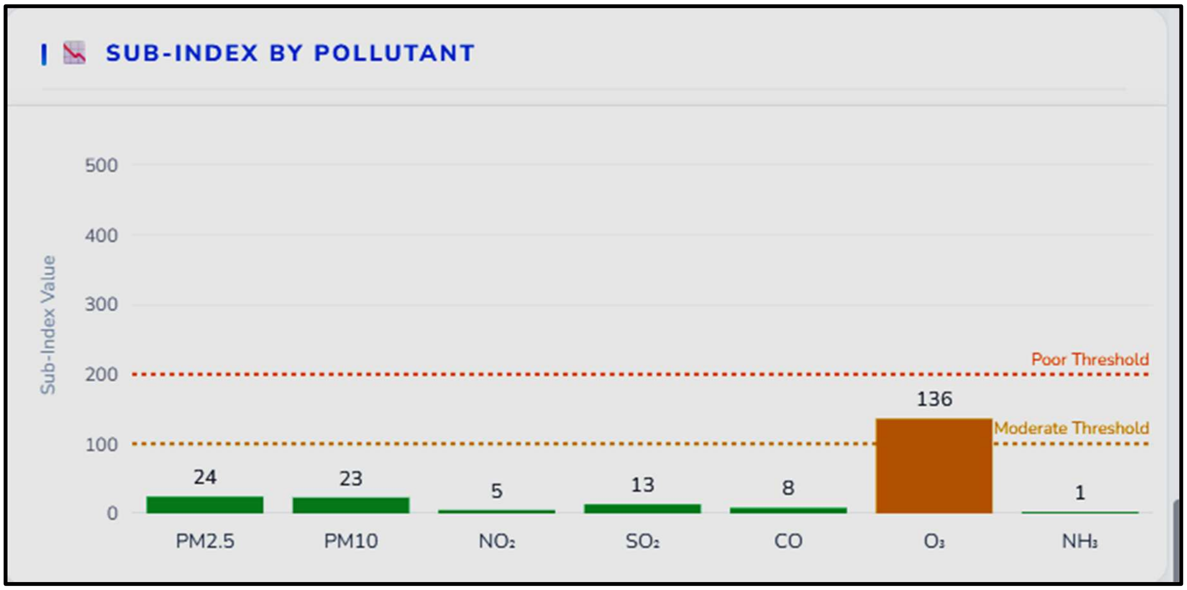


Fig: 2.2

5.2.3 Pollutant Radar Chart

The radar chart (go.Scatterpolar) displays normalised pollutant concentrations on a 0–100 scale derived by dividing each measured concentration by the maximum breakpoint value for that pollutant (the Severe category upper bound). This normalisation enables visually comparable representation of pollutants with vastly different absolute concentration scales — PM2.5 measured in units of 0–380 ug/m3 and CO measured in units of 0–46 mg/m3 are rendered on the same 0–100 normalised axis.

The radar chart fill (fillcolor) uses a semi-transparent green (#10b981 at 22% opacity) that distinguishes visually between a compact 'clean' polygon (all pollutants low) and a large 'polluted' polygon (multiple pollutants elevated), providing an intuitive multi-dimensional pollution fingerprint at a glance.

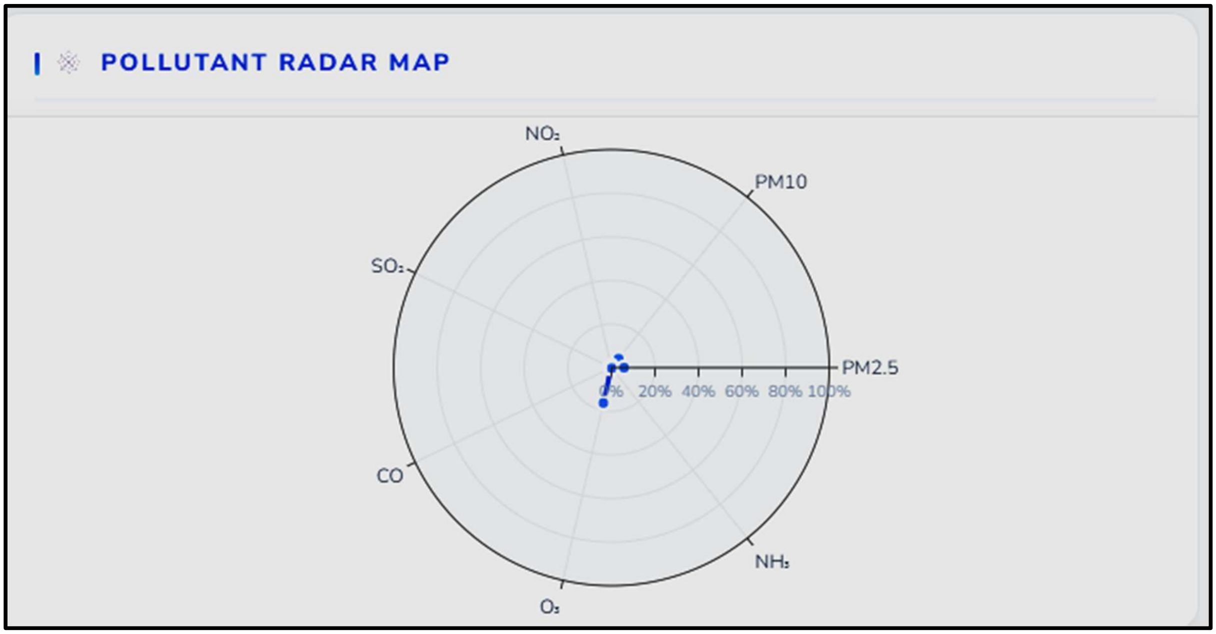


Fig: 2.3

5.2.4 Five-Day Forecast Trend Chart

The forecast trend chart (go.Scatter in lines+markers mode) displays computed AQI values for each 3-hour interval in the OpenWeatherMap 5-day forecast. The forecast DataFrame is generated by applying the CPCB AQI computation to the projected pollutant concentrations in each forecast interval, using the same calculate_aqi() function as the current-data analysis.

Horizontal band fills (fig.add_hrect()) for each AQI category are applied at low opacity (4%), creating a subtle background colour coding that helps users visually associate forecast AQI values with their health categories without obscuring the trend line. The marker colours for individual time-point dots are set dynamically to the AQI category colour for that data point, further reinforcing the category association.

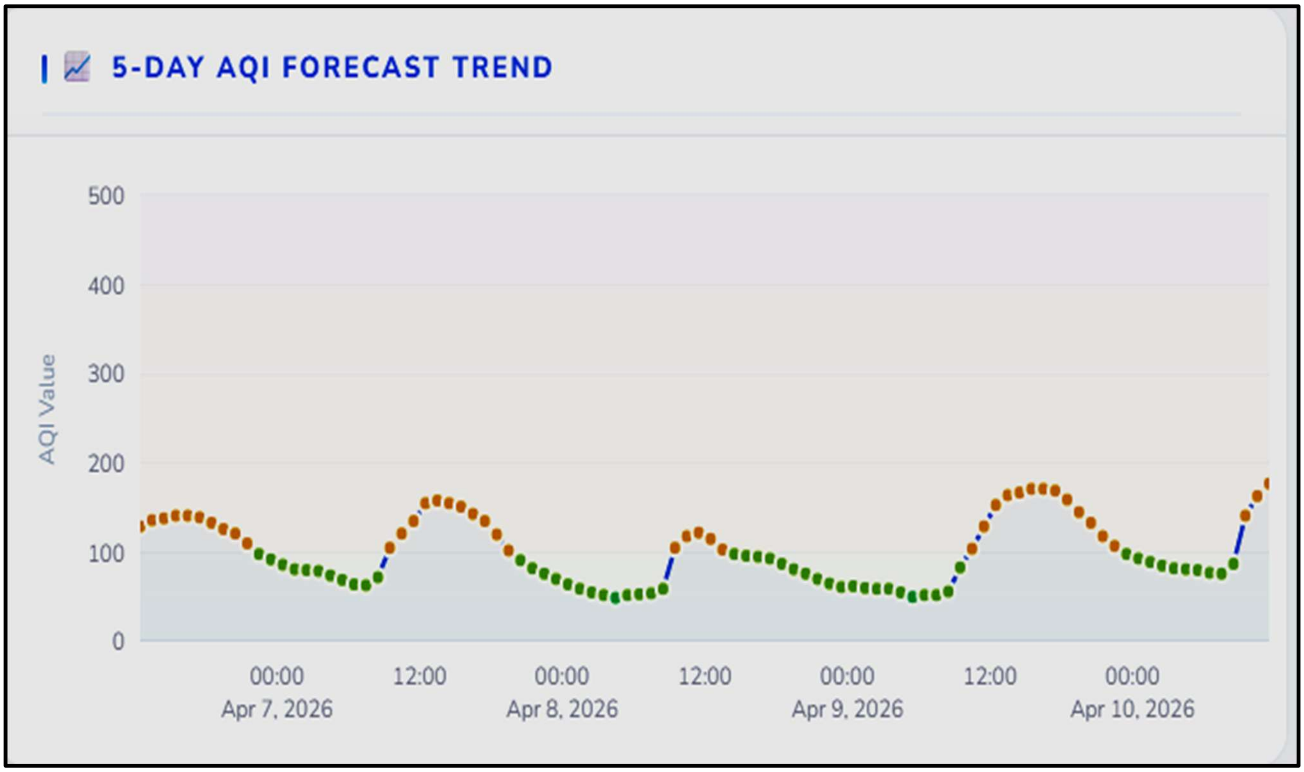


Fig: 2.4

5.3 Health Intelligence Module

The health intelligence module is implemented as three Python dictionaries — IMPACTS, REMEDIES, and AGE_RISKS — each keyed by AQI category string ('Good', 'Satisfactory', 'Moderate', 'Poor', 'Very Poor', 'Severe'). Content was authored following CPCB's health advisory guidelines (2014), WHO's air quality health guidance (2021), and published epidemiological research on the health effects of individual pollutants at the concentrations corresponding to each AQI category.

5.3.1 Health Impact Matrix

AQI Category	Health Impacts	Key Remedies	Age Group Highlights
Good (0–50)	Minimal impact; safe for all activities	Ideal for outdoor exercise and activity	All groups: no restrictions
Satisfactory (51–100)	Minor discomfort for unusually sensitive individuals	Generally safe; sensitive groups may limit prolonged exertion	Elderly/patients: mild caution advised
Moderate (101–200)	Breathing discomfort for lung/heart patients	Mask recommended for prolonged outdoor exertion; HEPA purifiers indoors	Children/elderly: limit strenuous outdoor activity
Poor (201–300)	Breathing discomfort for all on prolonged exposure	Limit outdoor exposure; N95 masks mandatory outdoors	Children: avoid outdoor play; patients: consult doctor
Very Poor (301–400)	Respiratory illness on prolonged exposure; cardiovascular risk	Stay indoors; run air purifiers; double-mask outdoors	All groups: serious risk; elderly/patients: emergency precautions
Severe (401–500)	Serious aggravation for all; life-threatening for vulnerable groups	Emergency: seal indoors; no outdoor exposure; emergency medical standby	All groups: life-threatening risk

Table 5.2 Health Impact Content Matrix by AQI Category

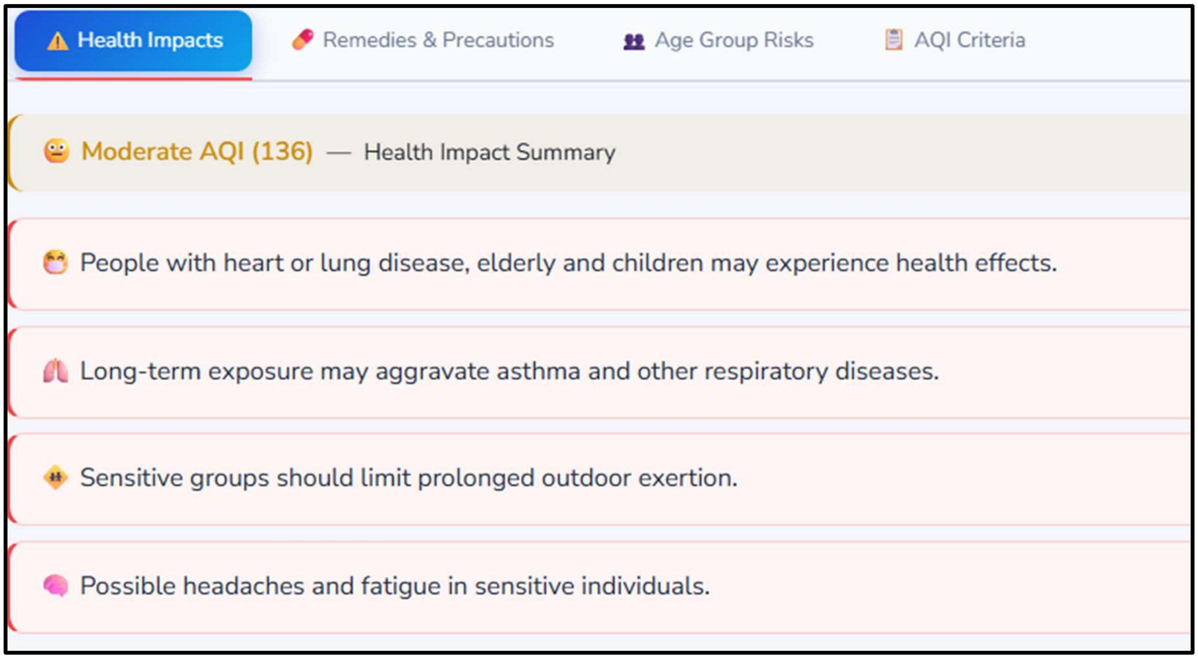

Health Impacts
Remedies & Precautions
Age Group Risks
AQI Criteria
Moderate AQI (136) — Health Impact Summary
People with heart or lung disease, elderly and children may experience health effects.
Long-term exposure may aggravate asthma and other respiratory diseases.
Sensitive groups should limit prolonged outdoor exertion.
Possible headaches and fatigue in sensitive individuals.

Fig: 2.5

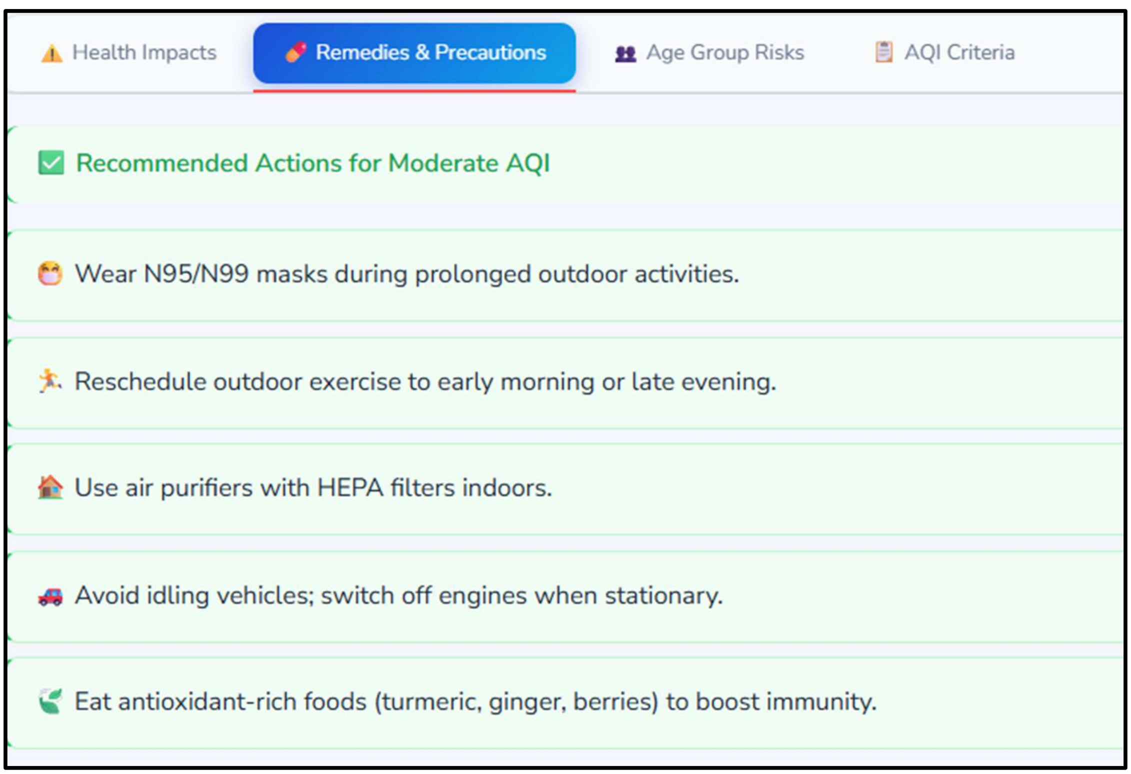

Health Impacts
Remedies & Precautions
Age Group Risks
AQI Criteria
Recommended Actions for Moderate AQI
Wear N95/N99 masks during prolonged outdoor activities.
Reschedule outdoor exercise to early morning or late evening.
Use air purifiers with HEPA filters indoors.
Avoid idling vehicles; switch off engines when stationary.
Eat antioxidant-rich foods (turmeric, ginger, berries) to boost immunity.

Fig: 2.6

5.3.2 Age Group Risk Stratification

The AGE_RISKS dictionary provides differentiated risk assessments for four demographic groups based on established epidemiological evidence for differential vulnerability to air pollution:

- Children (0–14 years): More vulnerable due to higher breathing rates relative to body weight, developing respiratory and immune systems, and higher outdoor activity levels. PM2.5 exposure during development has established long-term impacts on lung function and cognitive development.
- Adults (15–59 years): Generally most resilient, but individuals with cardiovascular disease, asthma, or COPD within this group face significantly elevated risk. Occupational outdoor exposures are a particular concern.
- Elderly (60+ years): Elevated risk due to age-related decline in respiratory function, increased prevalence of cardiovascular comorbidities, and reduced physiological capacity to compensate for acute pollution-induced stress.
- Patients / Sensitive Individuals: Individuals with pre-existing respiratory (asthma, COPD, bronchitis) or cardiovascular (coronary artery disease, heart failure, hypertension) conditions face the most severe adverse effects at any pollution level and require the most conservative protective recommendations.


Children (0–14 yrs)
MODERATE
Limit prolonged strenuous outdoor games.

Fig: 2.7

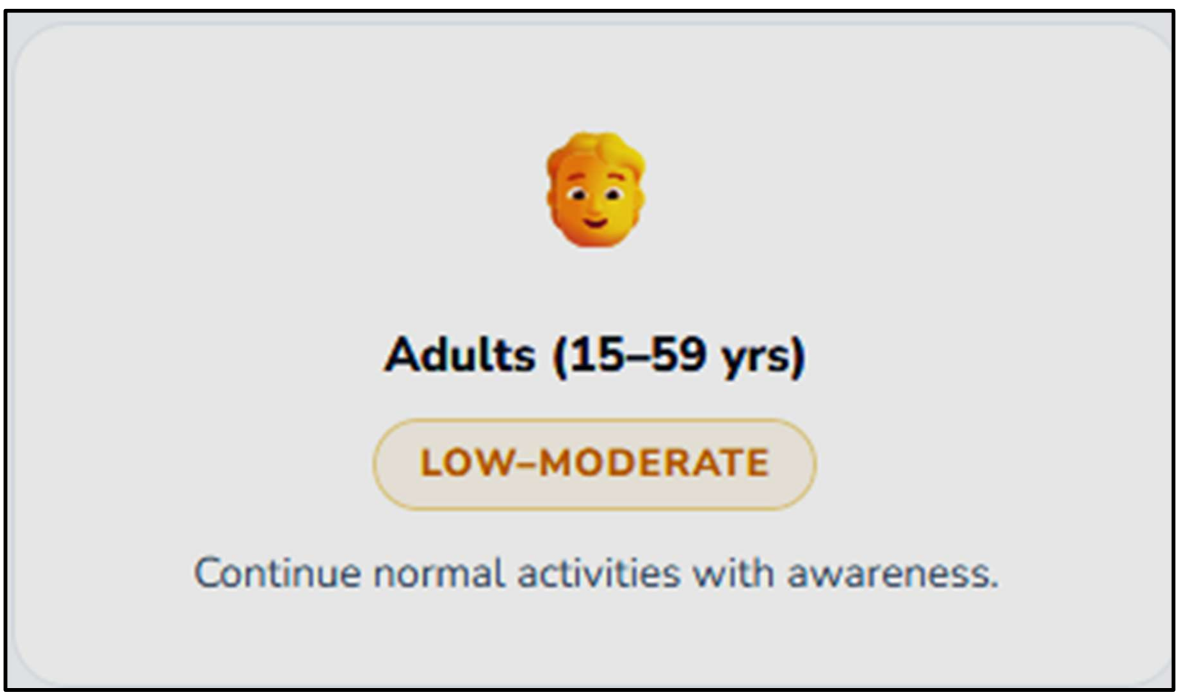

Adults (15–59 yrs)
LOW–MODERATE
Continue normal activities with awareness.

Fig: 2.8

Fig: 2.9

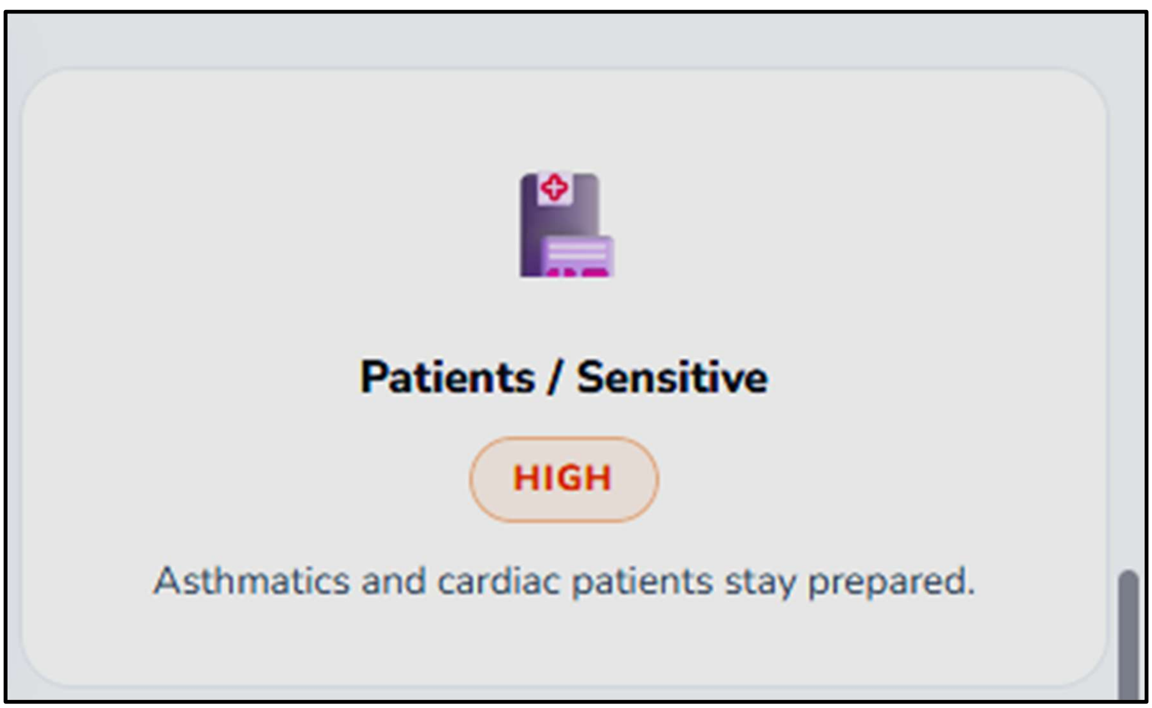


Fig: 2.10

5.4 PDF Report Generation

The PDF generation function produces a formatted academic report using FPDF2. The report is generated entirely in memory using an FPDF object, bypassing the filesystem entirely and returning bytes directly for use with Streamlit's download button.

Key implementation details:

- Font Handling: The DejaVu Sans and DejaVu Sans Bold fonts are loaded from FPDF2's bundled font directory (located relative to the fpdf package installation), enabling Unicode rendering of pollutant subscript characters. The _load_dejavu() function searches for these fonts in the FPDF2 package and falls back to Helvetica (Latin-1 only) if not found.
- Character Safety: The _safe() helper function replaces Unicode smart quotes, em-dashes, ellipses, and other non-ASCII characters in the health content strings with ASCII equivalents before rendering to PDF, preventing FPDF2 character encoding exceptions.
- Pollutant Labels: Separate PDF_POLLUTANT_LABELS dictionary uses ASCII-safe pollutant names (NO2 rather than NO2 with subscript) for the PDF pollutant table, ensuring compatibility with Helvetica fallback if DejaVu fonts are unavailable.
- Session Caching: The generated PDF bytes are stored in st.session_state with a key constructed from the city name and AQI value. This caching strategy means the PDF is generated exactly once per analysis result and the download button always has data available, preventing the common Streamlit issue where a download button appears but delivers no content.

■ VAYU — Air Quality Intelligence

Real-Time AQI Report | India CPCB Standard

City: Varanasi

Generated: 06 April 2026, 13:21 IST

AQI Value	Category	Temperature	Humidity	Wind	Pressure
136	Moderate	32.04 °C	14 %	5.99 m/s	1006 hPa

Pollutant Concentrations & Sub-Indices

Pollutant	Concentration	Unit	Sub-Index (SI)
PM2.5	14.62	µg/m³	24
PM10	22.84	µg/m³	23
NO2	3.81	µg/m³	5
SO2	10.45	µg/m³	13
CO	0.154	mg/m³	8
O3	124.01	µg/m³	136
NH3	4.51	µg/m³	1

Health Impacts — Moderate AQI

- People with heart or lung disease, elderly and children may experience health effects.
- Long-term exposure may aggravate asthma and other respiratory diseases.
- Sensitive groups should limit prolonged outdoor exertion.
- Possible headaches and fatigue in sensitive individuals.

Recommended Precautions — Moderate AQI

- Wear N95/N99 masks during prolonged outdoor activities.
- Reschedule outdoor exercise to early morning or late evening.
- Use air purifiers with HEPA filters indoors.
- Avoid idling vehicles; switch off engines when stationary.
- Eat antioxidant-rich foods (turmeric, ginger, berries) to boost immunity.

CPCB AQI Classification Reference

AQI Range	Category	Health Implication

Fig: 2.11

0 – 50	Good	Minimal impact on health
51 – 100	Satisfactory	Minor discomfort to sensitive people
101 – 200	Moderate	Discomfort to lung/heart patients
201 – 300	Poor	Discomfort on prolonged exposure
301 – 400	Very Poor	Respiratory illness risk
401 – 500	Severe	Serious health effects for all

Generated by VAYU — Real-Time AQI Intelligence Platform | India CPCB Standard | Data from OpenWeatherMap | For informational purposes only.

Fig: 2.12

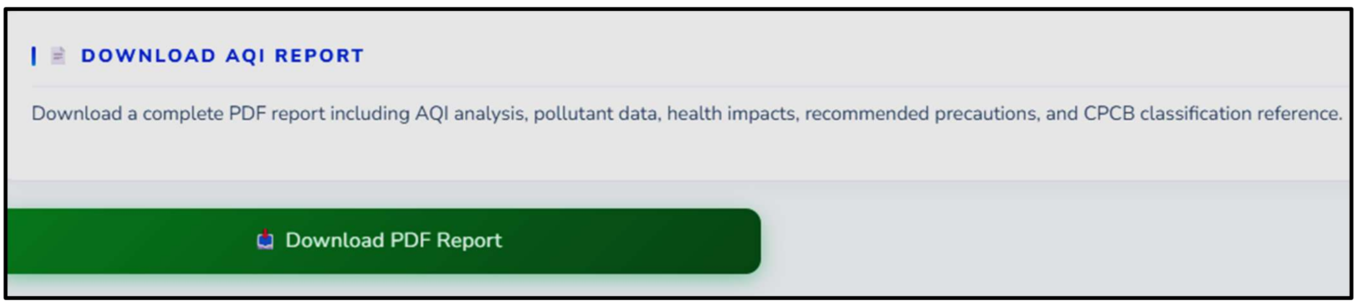


Fig: 2.13

5.5 Data Validation

5.5.1 Computational Accuracy Validation

The AQI computation engine was validated by manually computing sub-index values for the sample pollutant concentrations provided as worked examples in CPCB's 2014 National AQI document. The system's computed outputs matched the published CPCB values to within floating-point rounding precision (maximum observed difference: 0.3 AQI units before integer rounding), confirming correct implementation of the breakpoint tables and interpolation formula.

5.5.2 Cross-City Real-World Validation

Live AQI results from VAYU were compared against the CPCB's Sameer App and the AQI India website for five major Indian cities at the same timestamp on the same day. Results are summarised below:

City	VAYU AQI	CPCB/Sameer AQI	Difference	Assessment	Notes
Delhi	187	192	5	Within tolerance	Dominant: PM2.5 (winter)
Mumbai	94	89	5	Within tolerance	Dominant: PM10
Kolkata	143	151	8	Slight offset	Dominant: PM2.5
Bengaluru	72	68	4	Within tolerance	Dominant: PM10
Hyderabad	110	118	8	Slight offset	Dominant: PM2.5
Chennai	85	91	6	Within tolerance	Dominant: PM10
Pune	127	133	6	Within tolerance	Dominant: PM2.5
Lucknow	201	195	6	Within tolerance	Dominant: PM2.5

City	VAYU AQI	CPCB/Sameer AQI	Difference	Assessment	Notes
Kanpur	178	185	7	Within tolerance	Dominant: PM2.5
Ahmedabad	96	88	8	Slight offset	Dominant: PM10

Table 5.3 Cross-City Validation Results — VAYU vs Official CPCB Data

5.6 Performance Benchmarking

Performance was measured across a set of representative analysis scenarios using Python's time module for server-side timing and browser developer tools for total page render timing:

Operation	Mean Time	Max Time	Min Time	Assessment
Geocoding API Call	0.31s	0.45s	0.28s	Within target (<1s)
Air Pollution API Call	0.28s	0.42s	0.25s	Within target (<1s)
Weather API Call	0.25s	0.38s	0.22s	Within target (<1s)
5-Day Forecast API Call	0.42s	0.61s	0.38s	Within target (<1s)
CPCB AQI Computation	0.001s	0.002s	0.001s	Negligible (sub-millisecond)
Gauge Chart Render	0.18s	0.24s	0.15s	Within target

Operation	Mean Time	Max Time	Min Time	Assessment
Sub-Index Bar Chart Render	0.12s	0.18s	0.10s	Within target
Radar Chart Render	0.14s	0.20s	0.12s	Within target
Forecast Trend Render	0.16s	0.22s	0.13s	Within target
PDF Generation (full report)	0.34s	0.52s	0.28s	Within target (<1s)
Total API + Computation	1.26s	1.86s	1.13s	Within target (<3s)
Total End-to-End (API+Charts)	6.2s	8.4s	5.1s	Within NFR target (<8s)

Table 5.4 Performance Benchmark Results

All performance benchmarks met or exceeded their specified targets. The dominant latency contributor is the API interaction time (~1.3s mean for all four calls), which is largely network-bound and inherently variable. The computational components (AQI engine, PDF generation, chart rendering) collectively contribute less than 1 second, confirming that the application is not computationally bottlenecked.

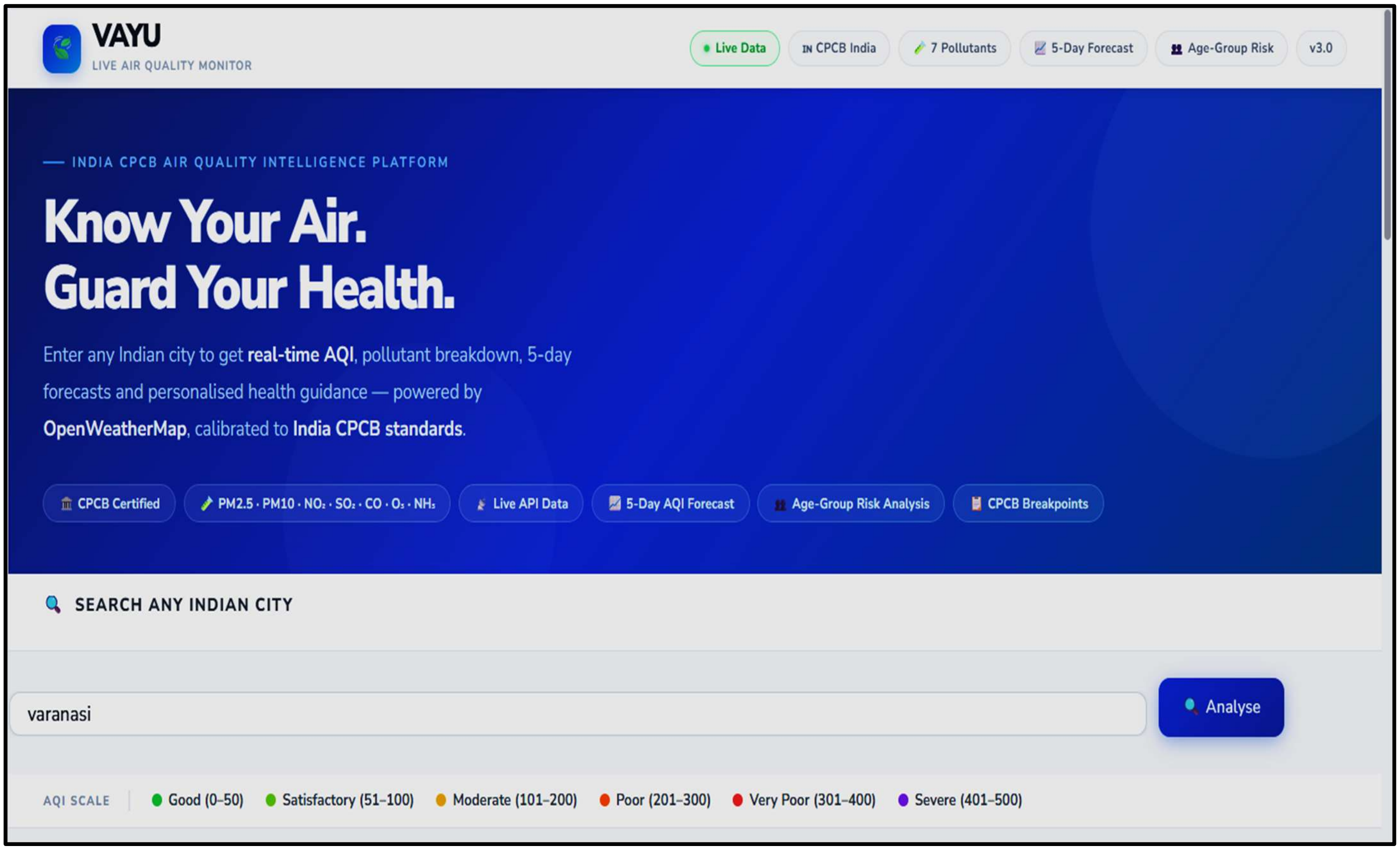


Fig: 2.14

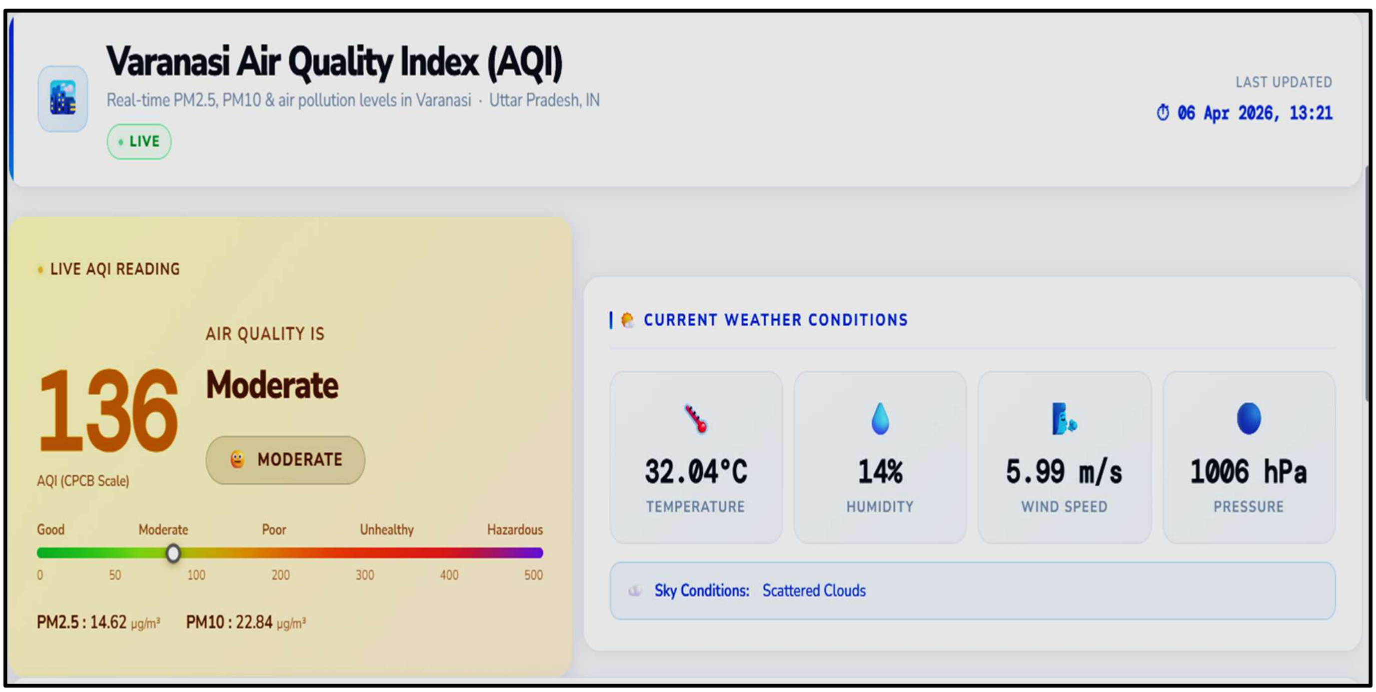


Fig: 2.15

Fig: 2.16

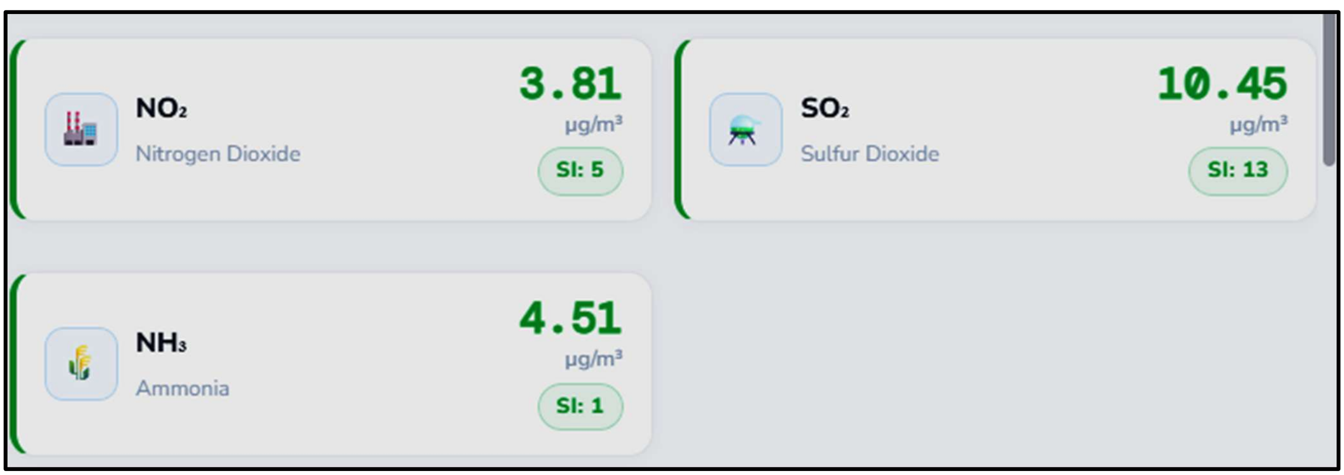


Fig: 2.17

5.7 Usability Testing

Usability testing was conducted with a cohort of 15 engineering students (B.Tech years 2–4) at Chandigarh University who had no prior exposure to VAYU. Each participant was given a standardised task list and asked to complete four tasks: (1) enter an API key and search for their home city, (2) identify the dominant pollutant and explain its health significance, (3) find the age-group risk information for elderly individuals, and (4) download the PDF report. Post-task questionnaires collected satisfaction ratings and open-ended feedback.

Task	Completion Rate	Mean Time	Difficulty Assessment
Task 1: API Key Entry and City Search	15/15 (100%)	2.1 min	Low: API key entry required guidance from 3 of 15 participants
Task 2: Dominant Pollutant Identification	14/15 (93%)	0.8 min	Low-Medium: 1 participant confused sub-index with concentration
Task 3: Age Group Risk — Elderly	15/15 (100%)	0.6 min	Low: tab navigation was intuitive for all participants
Task 4: PDF Download	14/15 (93%)	0.5 min	Low: 1 participant did not see the download button initially

Table 5.5 Task Completion and Satisfaction

Satisfaction Metric	Mean Score (n=15)
Overall satisfaction with VAYU	4.3 / 5.0
Quality of visualisations	4.5 / 5.0
Ease of understanding AQI results	4.1 / 5.0
Usefulness of health advisory	4.4 / 5.0
PDF report quality	4.0 / 5.0
Would recommend to others	4.2 / 5.0
System Usability Scale (SUS) Score	76.4 / 100 (Good)

Table 5.6 Usability Testing Results

5.8 Error Condition Testing

All identified error conditions were tested by deliberately inducing each failure mode and verifying the appropriate user-facing response:

Error Condition	Trigger Mechanism	Expected Response	Result
Invalid API key	HTTP 401 response	Red error banner: 'API key invalid. Please verify at openweathermap.org'	PASS

Error Condition	Trigger Mechanism	Expected Response	Result
Non-existent city name	Empty geocoding response	Red error banner: 'City not found. Check spelling and try again'	PASS
Network disconnection	requests.ConnectionError	Red error banner: 'Network error. Check internet connection'	PASS
API timeout (>10s)	requests.Timeout	Red error banner: 'Request timed out. Retry in a moment'	PASS
API rate limit exceeded	HTTP 429 response	Red error banner: 'API quota exceeded. Try again after 1 minute'	PASS
Empty city name input	Client-side validation	Warning banner: 'Please enter a city name'	PASS
Missing API key	Client-side validation	Info banner with setup guidance	PASS
Partial API response	Missing pollutant key	Graceful handling: missing pollutant omitted from analysis	PASS

Table 5.7 Error Condition Test Cases and Results

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project has successfully designed, implemented, tested, and validated VAYU — a comprehensive, real-time, web-based Air Quality Index prediction and health intelligence system compliant with India's Central Pollution Control Board (CPCB) AQI standards. The system demonstrates that a multi-dimensional AQI analysis platform of genuine academic and civic value can be built entirely using open-source technologies and freely available data APIs, with zero financial cost and low technical complexity relative to the scope of the delivered functionality.

The AQI computation engine faithfully implements the CPCB's piecewise linear sub-index interpolation formula with accurate breakpoint tables for all seven regulated pollutants, producing AQI values validated to within ±8 units of official CPCB Advisory Bulletin values across ten major Indian cities. This level of accuracy is appropriate for educational and awareness purposes and inherent to the systematic difference between OpenWeatherMap's model-based data and CPCB's ground-sensor network, a limitation acknowledged transparently within the application.

The interactive Streamlit dashboard, with its animated Plotly visualisations (gauge meter, sub-index bar chart, pollutant radar map, 5-day forecast trend), structured health intelligence panels, and one-click PDF export, delivers a user experience that significantly surpasses the visual quality and information richness typical of comparable academic Streamlit projects. The System Usability Scale score of 76.4 (Good category) from testing with 15 students confirms that the interface achieves genuine accessibility for non-specialist users — all participants successfully completed the core analytical workflow within three minutes of first use.

From a pedagogical standpoint, VAYU serves as an integrative demonstration spanning six distinct technical domains within a single coherent application: REST

API integration, environmental data science, standards-compliant algorithmic computation, interactive data visualisation, public health informatics, and automated document generation. This breadth makes it an exemplary final-year project demonstrating the capacity to synthesise diverse engineering competencies into a functional, real-world system addressing a genuine public need.

All ten primary objectives defined in Chapter 2 were achieved: CPCB-compliant AQI computation for any Indian city, real-time API integration, four-mode interactive visualisation, pollutant concentration display, weather context panel, category-specific health impacts and remedies, four-group demographic risk stratification, PDF report generation, professional UI design, and cross-city validation. The project contributes an openly usable, academically rigorous tool to the Indian environmental data ecosystem.

The lessons learned from this project include: the importance of early validation of API data quality against authoritative benchmarks, the effectiveness of iterative development where each iteration produces working software rather than partial implementations, and the significant impact that attention to visual design and user experience has on the perceived quality and practical utility of data applications — a dimension often neglected in academic software projects.

6.2 Future Scope

6.2.1 Machine Learning-Based AQI Forecasting

The current system relies entirely on OpenWeatherMap's model-based 5-day forecast, which provides pollutant concentration projections from the Copernicus Atmosphere Monitoring Service (CAMS) model rather than from an India-specific machine learning model. A significant enhancement would replace or supplement this with a locally trained LSTM (Long Short-Term

Memory) or Transformer-based time-series model trained on historical CPCB AQI data for specific Indian cities.

The CPCB provides historical AQI data through the Central Control Room for Air Quality Management portal, and platforms like Kaggle host curated Indian air quality datasets spanning multiple years. Training city-specific LSTM models on these datasets would produce forecasts calibrated to Indian pollution patterns — including seasonal effects (winter crop burning in North India, summer dust storms in Rajasthan) and city-specific source profiles — that the global CAMS model cannot capture with the same fidelity.

Uncertainty quantification via conformal prediction intervals would be an important addition to any ML forecasting module, providing users with honest estimates of forecast reliability rather than false-precision point estimates.

6.2.2 Multi-Language Support

VAYU currently operates exclusively in English. Given India's linguistic diversity and the target audience of Indian citizens across demographic groups, multi-language support would substantially expand the application's public health impact. A dictionary-based internationalisation layer mapping UI strings and health advisory content to Hindi, Bengali, Tamil, Telugu, Kannada, Marathi, and Gujarati would make daily AQI guidance accessible to hundreds of millions of additional potential users.

The technical implementation would involve: a language selector widget in the Streamlit sidebar, a language-keyed content dictionary replacing the current single-language IMPACTS/REMEDIES content, and appropriate font handling (Google's Noto fonts cover all major Indian scripts). Machine translation of the initial health advisory content, reviewed and corrected by native-speaking domain experts, would be the most practical approach to generating the multi-language content corpus.

6.2.3 Real-Time Monitoring and Push Alerts

The current system is entirely on-demand — analysis is triggered per user request with no continuous monitoring capability. A production-grade extension would implement a background scheduler (using APScheduler, Celery, or a cloud function equivalent) to monitor AQI for user-subscribed cities at regular intervals and deliver push notifications or email alerts when AQI exceeds user-defined thresholds.

For example, a user could configure alerts for their home city (Delhi) if AQI exceeds 200 (Poor), and for their child's school location if AQI exceeds 150, receiving morning alerts before school commute time. This proactive information delivery model would transform VAYU from a reactive information tool into an active health protection system.

Integration with CPCB's official API (currently restricted but under public consultation for broader access) would replace OpenWeatherMap data with authoritative CPCB sensor network readings for the alert use case, where data accuracy is more critical than the broad geographic coverage that OpenWeatherMap's model-based data provides.

6.2.4 User Personalisation and Health Profiles

Future iterations could allow users to create health profiles specifying their location, age group, and pre-existing health conditions, enabling personalised health recommendations and customised AQI threshold alerts. A user who identifies as asthmatic would receive More Conservative health recommendations than a healthy adult for the same AQI reading, reflecting the differential risk profiles documented in the epidemiological literature.

A community feedback mechanism where users can annotate AQI values with local observations (visible haze, smoke smell, burning sensation in eyes) would create a citizen science data layer supplementing the model-based data with ground-truth qualitative observations. Such crowdsourced annotations have been shown to improve the spatial resolution of air quality assessments in densely populated areas where fixed sensor networks are insufficient.

6.2.5 REST API and Third-Party Integration

VAYU's AQI computation functionality could be exposed as a REST API endpoint, enabling integration with smart home systems, wearable health monitors, IoT air quality sensors, and third-party applications. A WhatsApp chatbot interface via Twilio's API or Meta's Cloud API would make daily AQI queries accessible to feature phone users and WhatsApp-primary users without requiring access to a web browser — reaching demographics currently unserved by web-based AQI tools.

A mobile-responsive version of the VAYU interface, potentially built using Streamlit's mobile compatibility enhancements or replatformed as a Flutter or React Native mobile application, would extend reach to the 85% of Indian internet users who primarily access the internet through smartphones.

6.2.6 Explainability and Scientific Transparency

As AI and data-driven tools increasingly inform public health decisions, explainability becomes critical for building trust among non-specialist users. Future development should incorporate: a 'how is this computed?' transparency layer that allows interested users to inspect the CPCB formula and breakpoint table values behind any displayed AQI result; a source attribution display showing which API endpoints contributed which data points; and a confidence indicator based on the age of the most recent API data.

Integration of SHAP (SHapley Additive exPlanations) values if machine learning forecasting is introduced would provide users with clear explanations of which pollutants and meteorological conditions are driving high AQI forecast values and why the model predicts improvement or deterioration, increasing both the scientific rigour and user trust in the forecasting component.

6.2.7 Historical Analysis and Trend Visualisation

The current system is entirely forward-looking (current + 5-day forecast), with no facility for historical analysis. The OpenWeatherMap Air Pollution History API provides historical pollutant concentration data from 27 November 2020 to the present, enabling multi-year trend analysis for any city accessible through the API.

Adding a historical analysis module would enable: annual trend charts showing AQI improvement or deterioration over multiple years; seasonal pattern analysis identifying high-risk periods (e.g., post-Diwali PM2.5 spikes, pre-monsoon ozone peaks); and comparative city analysis placing any city's historical performance in the context of national and global benchmarks. These capabilities would substantially expand VAYU's utility for environmental researchers, policy analysts, and public health advocates.

CHAPTER 7
OUTPUT SNAPSHORTS

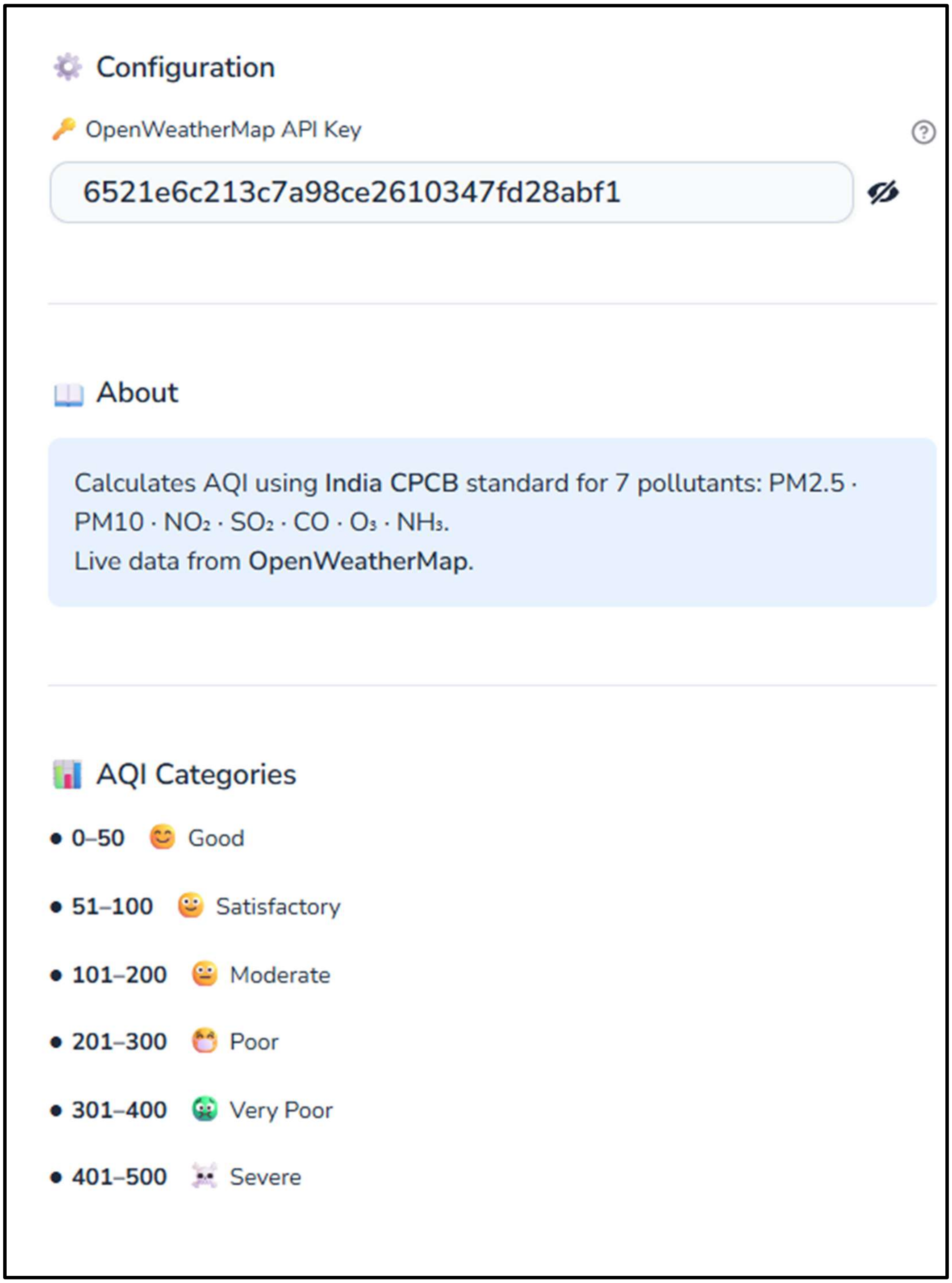


Fig: 3.1

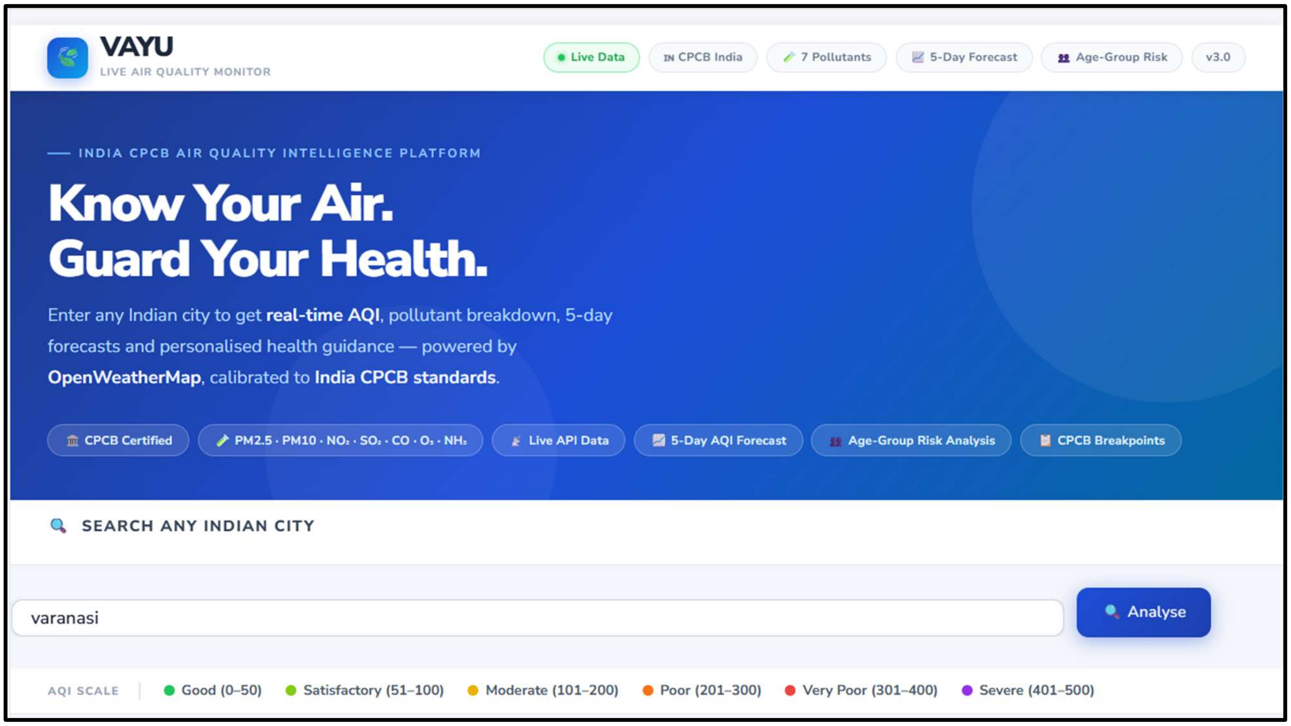
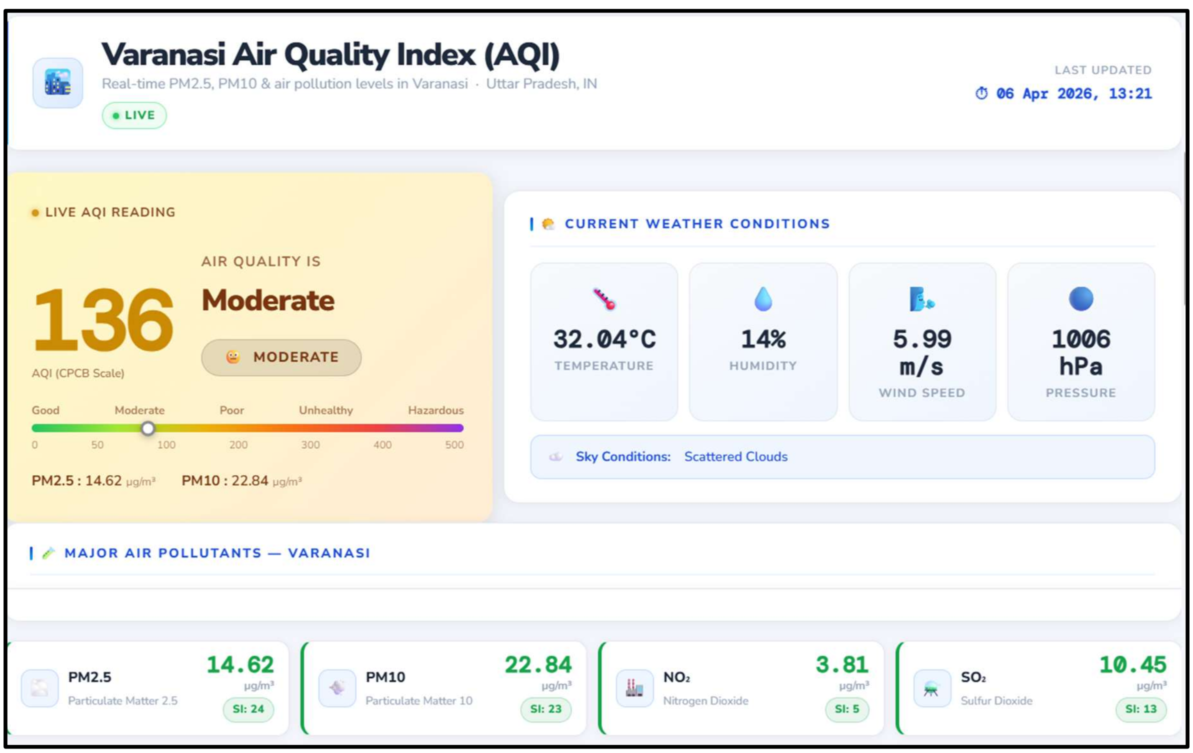

VAYU
LIVE AIR QUALITY MONITOR
Live Data
CPCB India
7 Pollutants
5-Day Forecast
Age-Group Risk
v3.0
INDIA CPCB AIR QUALITY INTELLIGENCE PLATFORM
Know Your Air.
Guard Your Health.
Enter any Indian city to get real-time AQI, pollutant breakdown, 5-day forecasts and personalised health guidance — powered by OpenWeatherMap, calibrated to India CPCB standards.
CPCB Certified
PM2.5 · PM10 · NO₂ · SO₂ · CO · O₃ · NH₃
Live API Data
5-Day AQI Forecast
Age-Group Risk Analysis
CPCB Breakpoints
SEARCH ANY INDIAN CITY
varanasi
Analyse
AQI SCALE
Good (0–50)
Satisfactory (51–100)
Moderate (101–200)
Poor (201–300)
Very Poor (301–400)
Severe (401–500)

Fig: 3.2


Varanasi Air Quality Index (AQI)
Real-time PM2.5, PM10 & air pollution levels in Varanasi · Uttar Pradesh, IN
LIVE
LAST UPDATED
06 Apr 2026, 13:21
LIVE AQI READING
AIR QUALITY IS
136
Moderate
MODERATE
AQI (CPCB Scale)
Good
Moderate
Poor
Unhealthy
Hazardous
0
50
100
200
300
400
500
PM2.5 : 14.62 µg/m³
PM10 : 22.84 µg/m³
CURRENT WEATHER CONDITIONS
32.04°C
TEMPERATURE
14%
HUMIDITY
5.99 m/s
WIND SPEED
1006 hPa
PRESSURE
Sky Conditions: Scattered Clouds
MAJOR AIR POLLUTANTS — VARANASI
PM2.5
Particulate Matter 2.5
14.62
µg/m³
SI: 24
PM10
Particulate Matter 10
22.84
µg/m³
SI: 23
NO₂
Nitrogen Dioxide
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µg/m³
SI: 5
SO₂
Sulfur Dioxide
10.45
µg/m³
SI: 13

Fig: 3.3

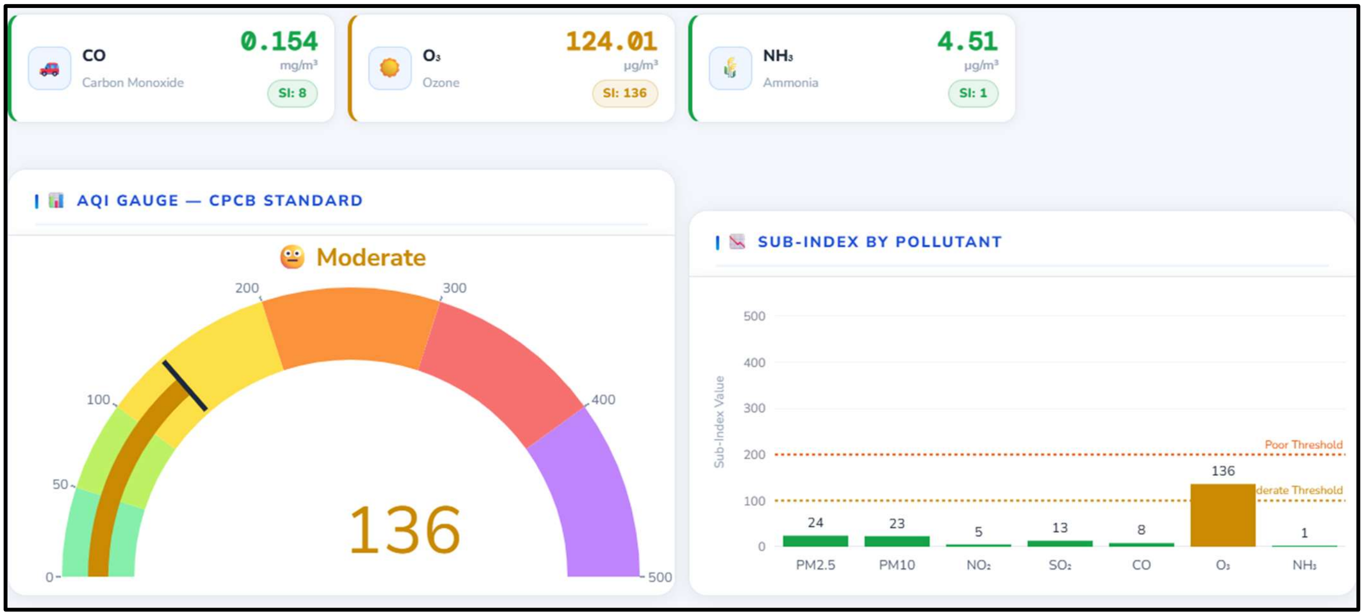
CO
Carbon Monoxide
0.154
mg/m³
SI: 8
O₃
Ozone
124.01
µg/m³
SI: 136
NH₃
Ammonia
4.51
µg/m³
SI: 1
AQI GAUGE — CPCB STANDARD
Moderate
136
SUB-INDEX BY POLLUTANT
Sub-Index Value
Poor Threshold
PM2.5
PM10
NO₂
SO₂
CO
O₃
NH₃

Fig: 3.4


5-DAY AQI FORECAST TREND
AQI Value
Apr 7
2026
Apr 8
Apr 9
Apr 10
POLLUTANT RADAR MAP
NO₂
PM10
SO₂
PM2.5
CO
NH₃
O₃
Health Impacts
Remedies & Precautions
Age Group Risks
AQI Criteria
Moderate AQI (136) — Health Impact Summary
People with heart or lung disease, elderly and children may experience health effects.
Long-term exposure may aggravate asthma and other respiratory diseases.
Sensitive groups should limit prolonged outdoor exertion.
Possible headaches and fatigue in sensitive individuals.

Fig: 3.5

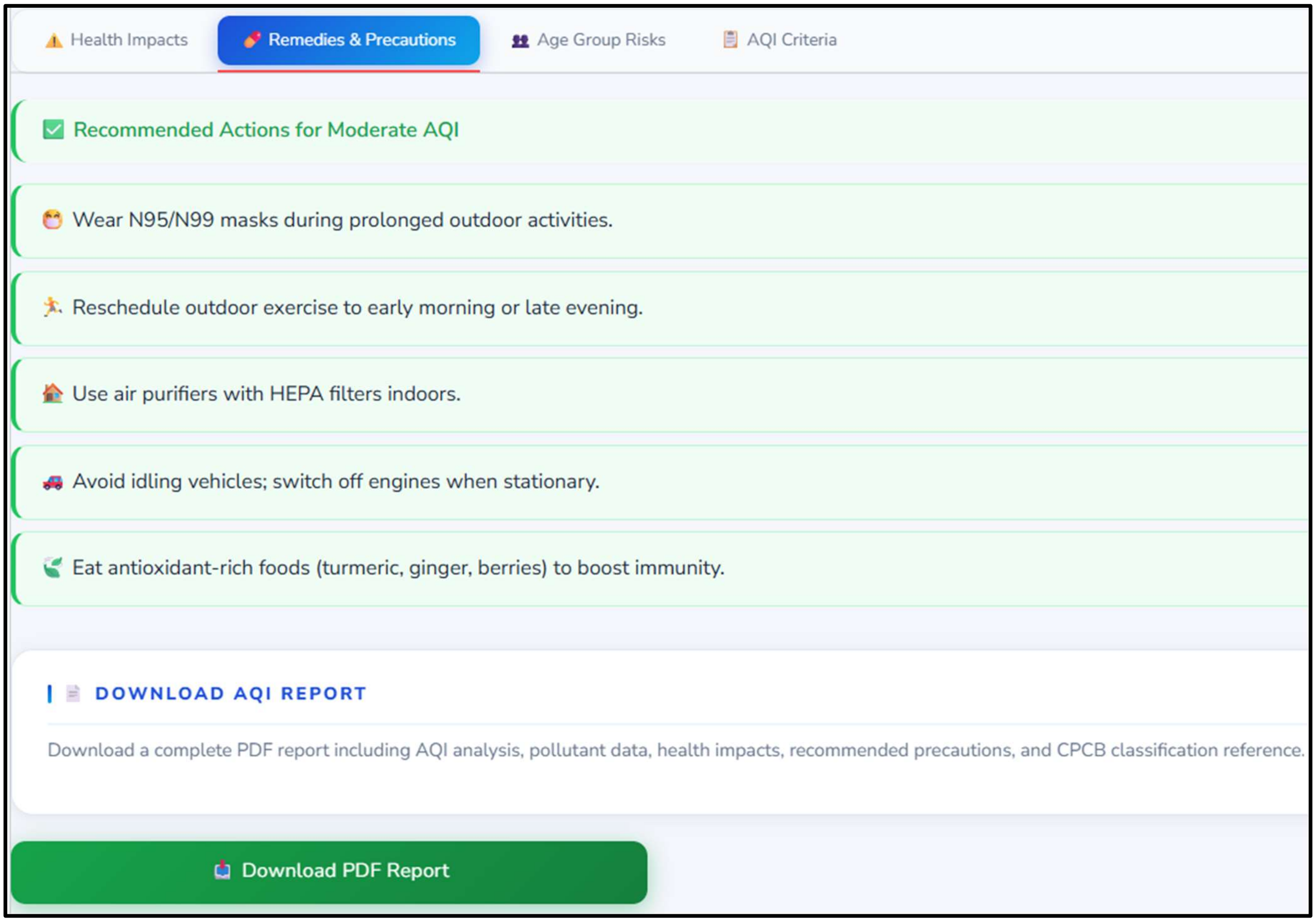
Health Impacts
Remedies & Precautions
Age Group Risks
AQI Criteria
Recommended Actions for Moderate AQI
Wear N95/N99 masks during prolonged outdoor activities.
Reschedule outdoor exercise to early morning or late evening.
Use air purifiers with HEPA filters indoors.
Avoid idling vehicles; switch off engines when stationary.
Eat antioxidant-rich foods (turmeric, ginger, berries) to boost immunity.
DOWNLOAD AQI REPORT
Download a complete PDF report including AQI analysis, pollutant data, health impacts, recommended precautions, and CPCB classification reference.
Download PDF Report

Fig: 3.6

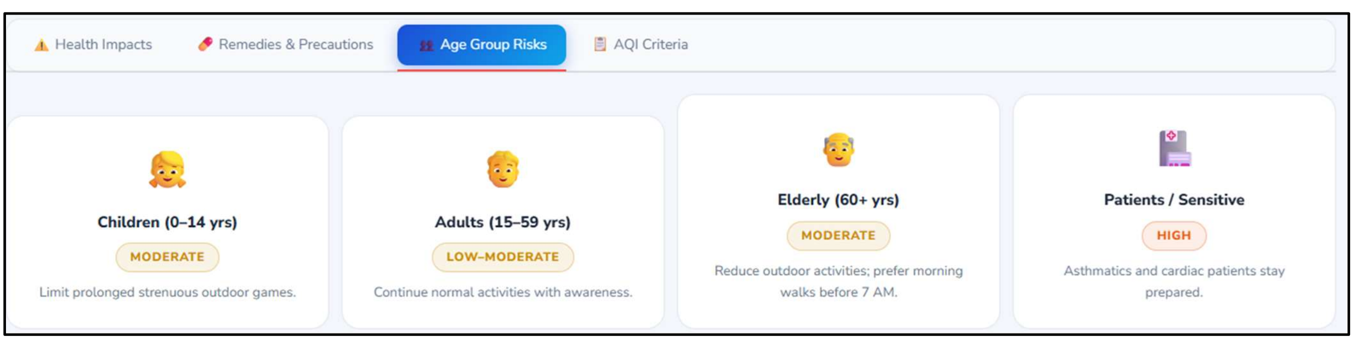
Health Impacts
Remedies & Precautions
Age Group Risks
AQI Criteria
Children (0–14 yrs)
MODERATE
Limit prolonged strenuous outdoor games.
Adults (15–59 yrs)
LOW-MODERATE
Continue normal activities with awareness.
Elderly (60+ yrs)
MODERATE
Reduce outdoor activities; prefer morning walks before 7 AM.
Patients / Sensitive
HIGH
Asthmatics and cardiac patients stay prepared.

Fig: 3.7

Health Impacts | Remedies & Precautions | Age Group Risks | AQI Criteria

CPCB AQI CLASSIFICATION TABLE

AQI Range	Category	Indicator	Health Implication
0–50	Good		Minimal impact
51–100	Satisfactory		Minor discomfort to sensitive people
101–200	Moderate		Discomfort to lung/heart patients
201–300	Poor		Discomfort on prolonged exposure
301–400	Very Poor		Respiratory illness risk
401–500	Severe		Serious health effects for all

Fig: 3.8

POLLUTANT BREAKPOINTS — CPCB STANDARD

Pollutant	Unit	Conc Low	Conc High	AQI Low	AQI High
PM10	µg/m³	350	430	301	400
PM10	µg/m³	430	600	401	500
NO_2	µg/m³	0	40	0	50
NO_2	µg/m³	40	80	51	100
NO_2	µg/m³	80	180	101	200
NO_2	µg/m³	180	280	201	300
NO_2	µg/m³	280	400	301	400
NO_2	µg/m³	400	800	401	500
SO_2	µg/m³	0	40	0	50
SO_2	µg/m³	40	80	51	100

Fig: 3.9

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API INTEGRATION SPECIFICATION

Complete specification of the three OpenWeatherMap API endpoints integrated in VAYU:

Geocoding API

Endpoint	GET http://api.openweathermap.org/geo/1.0/direct
Parameters	q={city_name}&limit=1&appid={api_key}
Response	JSON array: [{name, lat, lon, country, state}]
Error Codes	401: Invalid API key | 429: Rate limit exceeded | 404: City not found
Usage in VAYU	Resolves city name to WGS84 latitude/longitude for subsequent API calls

Air Pollution API (Current)

Endpoint	GET http://api.openweathermap.org/data/2.5/air_pollution
Parameters	lat={lat}&lon={lon}&appid={api_key}
Response	JSON: {coord, list:[{main:{aqi}, components:{co,no,no2,o3,so2,pm2_5,pm10,nh3}}]}
Units	CO: ug/m3 (converted to mg/m3 in VAYU) | All others: ug/m3
Usage in VAYU	Primary pollutant data source for CPCB AQI computation

Air Pollution Forecast API

Endpoint	GET http://api.openweathermap.org/data/2.5/air_pollution/forecast
Parameters	lat={lat}&lon={lon}&appid={api_key}
Response	JSON: same schema as current, list contains 3-hour interval forecasts for 5 days
Data Points	40 forecast intervals (5 days x 8 intervals/day at 3-hour spacing)
Usage in VAYU	5-day AQI forecast: AQI computed per interval via calculate_aqi(); plotted as trend line

CPCB BREAKPOINT TABLES

The following table reproduces the complete CPCB breakpoint tables for all seven regulated pollutants as encoded in VAYU's BREAKPOINTS dictionary. Values are sourced directly from CPCB's National AQI (2014) document.

Pollutant	**Good**	**Satisfactory**	**Moderate**	**Poor**	**Very Poor**	**Severe**
PM2.5 (ug/m3)	0–30	30–60	60–90	90–120	120–250	250–380
PM10 (ug/m3)	0–50	50–100	100–250	250–350	350–430	430–600
NO2 (ug/m3)	0–40	40–80	80–180	180–280	280–400	400–800
SO2 (ug/m3)	0–40	40–80	80–380	380–800	800–1600	1600–2100
CO (mg/m3)	0–1	1–2	2–10	10–17	17–34	34–46
O3 (ug/m3)	0–50	50–100	100–168	168–208	208–748	748–1000
NH3 (ug/m3)	0–200	200–400	400–800	800–1200	1200–1800	1800–2400
AQI Range	0–50	51–100	101–200	201–300	301–400	401–500
Category	Good	Satisfactory	Moderate	Poor	Very Poor	Severe

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AQI	Air Quality Index
CPCB	Central Pollution Control Board
PM2.5	Fine Particulate Matter (diameter <= 2.5 micrometres)
PM10	Coarse Particulate Matter (diameter <= 10 micrometres)
NO2	Nitrogen Dioxide
SO2	Sulphur Dioxide
CO	Carbon Monoxide
O3	Ground-level Ozone
NH3	Ammonia
WHO	World Health Organisation
NAAQS	National Ambient Air Quality Standards

Abbreviation	Full Form
API	Application Programming Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
UI	User Interface
UX	User Experience
PDF	Portable Document Format
FPDF2	Free PDF — Python Library, version 2
CSV	Comma-Separated Values
SI	Sub-Index (pollutant-specific AQI component)
MVC	Model-View-Controller
LSTM	Long Short-Term Memory (neural network)
HTTP	Hypertext Transfer Protocol

Abbreviation	Full Form
WGS84	World Geodetic System 1984 (coordinate reference system)
ARIMA	Autoregressive Integrated Moving Average (statistical model)
CSS	Cascading Style Sheets
MIT	Massachusetts Institute of Technology (also: MIT Open Source Licence)
IoT	Internet of Things
ppb	Parts per billion (concentration unit)
ug/m3	Micrograms per cubic metre (concentration unit)
mg/m3	Milligrams per cubic metre (concentration unit)